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United States Patent Application Publication

20250283711

Kind Code

A1

Publication Date

September 11, 2025

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SYSTEMS AND METHODS FOR MULTIPATH IMAGING

Abstract

A fiber optic imaging system generates multipath image data. The system comprises: an optical coherence tomography (OCT) system comprising an OCT fiber optically connected to an interferometric detector; a multi-clad fiber (MCF) for receiving sampled light that has interacted with a sample and propagating the sampled light as a fundamental MCF mode and one or more higher order MCF modes; and an optical joint for coupling the sampled light from the MCF into the OCT fiber, the optical joint configured to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber where the at least some light energy propagates as OCT return light.

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Family ID: 91969816

Appl. No.: 19/214633

Filed: May 21, 2025

Related U.S. Application Data

parent WO continuation PCT/CA2024/050069 20240122 PENDING child US 19214633
us-provisional-application US 63440954 20230125

Publication Classification

Int. Cl.: G01B9/02091 (20220101); A61B5/00 (20060101)

U.S. Cl.:

CPC G01B9/02091 (20130101); A61B5/0066 (20130101); A61B5/0084 (20130101); A61B2576/00 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of Patent Cooperation Treaty (PCT) application No. PCT/CA2024/050069 entitled SYSTEMS AND METHODS FOR

MULTIPATH IMAGING and having an international filing date of 22 Jan. 2024 which in turn claims priority from, and for the purposes of the United States of America the benefit under 35 U.S.C. § 119 of, U.S. application No. 63/440,954 filed 25 Jan. 2023 and entitled NOVEL METHODS FOR MULTIPATH CONTRAST IMAGING. All of the applications referred to in this paragraph are hereby incorporated herein by reference for all purposes.

FIELD

[0002] This invention relates to fiber optic based optical imaging, such as endoscopy, for example. Particular embodiments provide methods and systems for optical coherence tomography (OCT) imaging using multi-clad fibers and corresponding imaging techniques.

BACKGROUND

[0003] Optical endoscopes exploit the near-lossless transmission, narrow imaging cores and flexibility of fiber optics to provide high resolution, minimally invasive imaging in, for example, luminal organs deep within the body.

[0004] Low signal modalities such as contrast-free fluorescence imaging necessitate multi-mode fibers (MMF). MMF have high collection efficiencies from refractive index profiles (RIPs) with wide diameter and high numerical aperture (NA) cores.

[0005] Confocal endomicroscopy makes use of single-mode fibers (SMF) in place of a pinhole to maintain optical sectioning properties of a free-space system. Time-gated techniques such as OCT also require SMF to limit modal dispersion as additional modes introduce ghost images which can obscure the true image.

[0006] Optical fibers with more complex RIPs further enable more elaborate imaging schemes. For example, multi-type imaging comprising OCT and an additional imaging type makes use of double-clad fiber (DCF) which has a single-mode core for OCT, and a multimode inner-cladding for the additional imaging type. OCT provides volumetric imaging of subsurface tissue morphology; thus, functional imaging modalities which provide information regarding tissue biochemistry, such as autofluorescence imaging (AFI), may be used, for example, as the additional imaging type.

[0007] Utilizing complex RIP geometries to implement multi-type imaging in a single fiber (e.g. a DCF) typically has a cost. For instance, the doped cores of DCF generate additional background fluorescence when compared to a pure-silica MMF core, reducing fluorescence image quality. In the OCT type, imaging with DCF as opposed to SMF results in time shifted multipath artifacts, i.e., ghost images smeared in the A-line direction that may be superimposed on the fundamental image.

[0008] The distance (in the A-line direction) between the true image and any multipath artifacts is typically linearly related to the length of DCF used in the sample arm of the endoscopic system. In theory, the impacts of multipath artifacts could be significantly mitigated by using a long length of DCF (>5 m) in the sample arm. In practice, the length of DCF in the sample arm of an endoscopic imaging catheter is typically constrained to <2 m by mechanical and clinical needs; thus, in prior art techniques, these multipath artifact impacts are generally mitigated by folding the multipath artifacts over the zero-reference delay to prevent or mitigate superposition of the multipath artifacts over the fundamental image.

[0009] There is a general desire to provide systems and methods for OCT imaging that are capable of using a MCF (e.g. DCF) in the sample arm (e.g. to support other imaging modalities), while mitigating the effect of, or making use of, multipath artifacts.

[0010] There is a general desire for improved techniques for fiber optic based optical imaging.

SUMMARY

[0011] One aspect of the invention provides a fiber optic imaging system for generating multipath image data. The system comprises: an optical coherence tomography (OCT) system comprising an OCT fiber optically connected to an interferometric detector; a multi-clad fiber (MCF) for receiving sampled light that has interacted with a sample and propagating the sampled light as a fundamental MCF mode and one or more higher order MCF modes; an optical joint for coupling the sampled light from the MCF into the OCT fiber, the optical joint configured to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber where the at least some light energy propagates as OCT return light.

[0012] The imaging system may comprise a controller connected to the interferometric detector to receive, from the interferometric detector, OCT image data based on the OCT return light. The controller may be configured to: extract fundamental mode image data and higher order mode image data from the OCT image data, wherein the extraction is based on A-line coordinates of the OCT image data; and generate multipath image data (e.g. diagnostic image data) based on both of the fundamental mode image data and the higher

order mode image data.

[0013] The controller may be configured to: extract the fundamental mode image data based on light energy from the fundamental MCF mode that is coupled into the fundamental mode of the OCT fiber; and extract the higher order mode image data based on light energy from the one or more higher order MCF modes that is coupled into the fundamental mode of the OCT fiber.

[0014] The controller may be configured to extract the fundamental mode image data and the higher order mode image data from the OCT image data using upper and lower A-line coordinate thresholds.

[0015] The MCF may introduce a phase delay between the fundamental MCF mode and the one or more higher order MCF modes.

[0016] The controller may be configured to extract the fundamental mode image data and the higher order mode image data based on the phase delay between the fundamental MCF mode and the one or more higher order MCF modes.

[0017] An amount of the phase delay may depend on a length of the MCF. The amount of the phase delay may be sufficient to enable extraction of the fundamental mode image data and the higher order mode image data from the OCT image data based on the A-line coordinates of the OCT image data.

[0018] The MCF may comprise a MCF core and at least one light-transmitting MCF cladding. The fundamental MCF mode may propagate substantially in the MCF core and the one or more higher order MCF modes may propagate substantially in the at least one MCF cladding.

[0019] The fundamental MCF mode may comprise light energy received from relatively low numerical aperture portions of the sampled light and the one or more higher order MCF modes may comprise light energy received from relatively high numerical aperture portions of the sampled light.

[0020] The MCF may propagate incident light from the optical joint toward the sample in an incident direction opposite to a direction of propagation of the sampled light. At least a portion of the incident light may interact with the sample to become the sampled light.

[0021] The incident light may comprise: fundamental mode incident light that propagates in the incident direction in a fundamental mode of the MCF; and higher order mode incident light that propagates in the incident direction in one or more higher order modes of the MCF.

[0022] The one or more higher order MCF modes of the sampled light may comprise a first component corresponding to the fundamental mode incident light that has interacted with the sample and a second component corresponding to the higher order mode incident light that has interacted with the sample.

[0023] The controller may be configured to extract the higher order mode image data based on light energy from the first component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.

[0024] The controller may be configured to extract additional higher order mode image data from the OCT image data based on light energy from the second component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.

[0025] The controller may be configured to extract the additional higher order mode image data based on A-line coordinates of the OCT image data.

[0026] The controller may be configured to extract the fundamental mode image data, the higher order mode image data and the additional higher order mode image data based on differential phase delay between different propagation modes introduced in the MCF.

[0027] The controller may be configured to generate multipath image data based at least in part on the additional higher order mode image data.

[0028] The optical joint may be configured to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into the fundamental of the OCT fiber by introducing an offset between the MCF and the OCT fiber.

[0029] The offset may comprise a spatial offset between an axis of the MCF and an axis of the OCT fiber. The offset may comprise an angular offset between an axis of the MCF and an axis of the OCT fiber.

[0030] The optical joint may comprise one or more optical components (e.g. lenses and/or mirrors) which introduce optical offset between the MCF and the OCT fiber.

[0031] The optical joint may comprise a fiber optic rotary joint (FORJ) configured to rotate the MCF about its axis relative to the sample.

[0032] The fundamental mode image data and the higher order mode image data may comprise volumetric data.

[0033] The controller may be configured to: process the fundamental mode image data to obtain two-dimensional processed fundamental mode image data; process the higher order mode image data to obtain two-

dimensional processed higher order mode image data; and generate the multipath image data based on the two-dimensional processed fundamental mode image data and the two-dimensional processed higher order mode image data.

[0034] The controller may be configured to: process the fundamental mode image data to obtain the two-dimensional processed fundamental mode image data as a fundamental mode en face image by repeating: for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in the fundamental mode image data, aggregating (e.g. adding or averaging) pixel values along the A-line direction between lower and upper fundamental mode A-line coordinate thresholds to obtain one corresponding pixel value of the two-dimensional processed fundamental mode image data; and process the higher order mode image data to obtain the two-dimensional processed higher order mode image data as a higher order mode en face image by repeating: for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in the higher order mode image data, aggregating (e.g. adding or averaging) pixel values along the A-line direction between lower and upper higher order mode A-line coordinate thresholds to obtain one corresponding pixel value of the two-dimensional processed higher order mode image data.

[0035] The controller may be configured to generate the multipath image data based on a combination of the fundamental mode en face image and the higher order mode en face image.

[0036] The controller may be configured to generate the multipath image data based on a ratio of the fundamental mode en face image and the higher order mode en face image.

[0037] The MCF may comprise a double-cladded fiber (DCF).

[0038] The imaging system may comprise a second imaging type system, the second imaging type system configured to direct second incident light from the optical joint toward the sample by propagating the second incident light in the MCF, wherein at least a portion of the second incident light interacts with the sample to become second sampled light and wherein the second imaging type system is configured to collect the second sampled light which propagates back toward the optical joint in the MCF.

[0039] The imaging system may comprise a first light source that generates first incident light, at least a portion of which interacts with the sample to become the sampled light. The second imaging type system may comprise a second light source for generating the second incident light.

[0040] A wavelength of the first incident light may be different than a wavelength of the second incident light.

[0041] The optical joint may comprise a wavelength division multiplexer configured to combine the first incident light and the second incident light to generate combined incident light.

[0042] The optical joint may comprise a fiber optic rotary joint (FORJ) configured to rotate the MCF about its axis relative to the sample and for coupling the combined incident light into the MCF.

[0043] The optical joint may comprise a multi-clad fiber coupler (MCFC) configured to separate the sampled light from the second sampled light.

[0044] The second imaging type system may comprise an optical detector configured to detect at least a portion of the second sampled light. The controller may be configured to generate a second type image based on the detected portion of the second sampled light.

[0045] The second type image may comprise an en face image.

[0046] The second imaging type system may comprise an autofluorescence imaging (AFI) system.

[0047] Another aspect of the invention provides a method for generating multipath image data. The method comprises: providing a multi-clad fiber (MCF); receiving sampled light that has interacted with the sample and propagating the sampled light in the MCF as a fundamental MCF mode and one or more higher order MCF modes; coupling the sampled light from the MCF into an optical coherence tomography (OCT) fiber, wherein coupling the sampled light comprises coupling at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber, where at least some light energy propagates as OCT return light; generating OCT image data based on the OCT return light; extracting fundamental mode image data and higher order mode image data from the OCT image data, wherein the extracting is based on A-line coordinates of the OCT image data; and generating the multipath image data based on both of the fundamental mode image data and the higher order mode image data.

[0048] The fundamental mode image data may be based on light energy from the fundamental MCF mode that is coupled into the fundamental mode of the OCT fiber; and the higher order mode image data may be based on light energy from the one or more higher order MCF modes that is coupled into the fundamental mode of the OCT fiber.

[0049] Extracting the fundamental mode image data and higher order mode image data from the OCT image data may comprise using upper and lower A-line thresholds.

[0050] The MCF may introduce a phase delay between the fundamental MCF mode and the one or more

higher order MCF modes.

[0051] Extracting the fundamental mode image data and higher order mode image data from the OCT image data may be based on the phase delay between the MCF mode and the one or more higher order MCF modes.

[0052] An amount of the phase delay may depend on a length of the MCF. The amount of the phase delay may be sufficient to enable extraction of the fundamental mode image data and the higher order mode image data from the OCT image data based on the A-line coordinates of the OCT image data.

[0053] The MCF may comprise a MCF core and at least one light-transmitting MCF cladding. The fundamental MCF mode may propagate substantially in the MCF core and the one or more higher order MCF modes may propagate substantially in the at least one MCF cladding.

[0054] The fundamental MCF mode may comprise light energy received from relatively low numerical aperture portions of the sampled light and the one or more higher order MCF modes may comprise light energy received from relatively high numerical aperture portions of the sampled light.

[0055] The MCF may propagate incident light toward the sample in an incident direction opposite to a direction of propagation of the sampled light. At least a portion of the incident light may interact with the sample to become the sampled light.

[0056] The incident light may comprise: fundamental mode incident light that propagates in the incident direction in a fundamental mode of the MCF; and higher order mode incident light that propagates in the incident direction in one or more higher order modes of the MCF.

[0057] The one or more higher order MCF modes of the sampled light may comprise a first component corresponding to the fundamental mode incident light that has interacted with the sample and a second component corresponding to the higher order mode incident light that has interacted with the sample.

[0058] Extracting the higher order mode image data may be based on light energy from the first component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.

[0059] The method may comprise extracting additional higher order mode image data from the OCT image data based on light energy from the second component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.

[0060] The method may comprise extracting the additional higher order mode image data based on A-line coordinates of the OCT image data.

[0061] The method may comprise extracting the fundamental mode image data, the higher order mode image data and the additional higher order mode image data based on differential phase delay between different propagation modes introduced in the MCF.

[0062] Generating the multipath image data may be based at least in part on the additional higher order mode image data.

[0063] Coupling at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into the fundamental of the OCT fiber may comprise providing an offset between the MCF and the OCT fiber.

[0064] The offset may comprise a spatial offset between an axis of the MCF and an axis of the OCT fiber.

[0065] The offset may comprise an angular offset between an axis of the MCF and an axis of the OCT fiber.

[0066] Providing the offset may comprise locating one or more optical components (e.g. lenses and/or mirrors) in an optical path between the MCF and the OCT fiber to thereby introduce optical offset between the MCF and the OCT fiber.

[0067] Coupling at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into the fundamental of the OCT fiber may comprise locating a fiber optic rotary joint (FORJ) configured to rotate the MCF about its axis relative to the sample in an optical path between the MCF and the OCT fiber.

[0068] The fundamental mode image data and the higher order mode image data may comprise volumetric data.

[0069] The method may comprise: processing the fundamental mode image data to obtain two-dimensional processed fundamental mode image data; processing the higher order mode image data to obtain two-dimensional processed higher order mode image data; and generating the multipath image data based on the two-dimensional processed fundamental mode image data and the two-dimensional processed higher order mode image data.

[0070] Processing the fundamental mode image data to obtain the two-dimensional processed fundamental mode image data may comprise processing the fundamental mode image data to obtain a fundamental mode end face image by repeating: for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in

the fundamental mode image data, aggregating (e.g. adding or averaging) pixel values along the A-line direction between lower and upper fundamental mode A-line coordinate thresholds to obtain one corresponding pixel value of the two-dimensional processed fundamental mode image data. Processing the higher order mode image data to obtain the two-dimensional processed higher order mode image data may comprise processing the higher order mode image data to obtain a higher order mode en face image by repeating: for each for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in the higher order mode image data, aggregating (e.g. adding or averaging) pixel values along the A-line direction between lower and upper higher order mode A-line coordinate thresholds to obtain one corresponding pixel value of the two-dimensional processed higher order mode image data.

[0071] Generating the multipath image data may comprise generating the multipath image data based on a combination of the fundamental mode en face image and the higher order mode en face image.

[0072] Generating the multipath image data may comprise generating the multipath image data based on a ratio of the fundamental mode en face image and the higher order mode en face image.

[0073] The MCF may comprise a double-cladded fiber (DCF).

[0074] The method may comprise: providing a second imaging type system; propagating second incident light toward the sample by propagating the second incident light in the MCF; wherein at least a portion of the second incident light interacts with the sample to become second sampled light; and collecting the second sampled light which propagates back toward the second imaging type system in the MCF.

[0075] The method may comprise: generating first incident light by a first light source, at least a portion of which interacts with the sample to become the sampled light; and generating the second incident light by a second light source.

[0076] A wavelength of the first incident light may be different than a wavelength of the second incident light.

[0077] The method may comprise combining the first incident light and the second incident light to generate combined incident light.

[0078] The method may comprise coupling the combined incident light into the MCF by a fiber optic rotary joint (FORJ).

[0079] The method may comprise separating the sampled light from the second sampled light using a multi-clad fiber coupler (MCFC).

[0080] The method may comprise: detecting at least a portion of the second sampled light; and generating a second type image based on the detected portion of the second sampled light.

[0081] The second type image may comprise an en face image.

[0082] The second imaging type system may comprise an autofluorescence imaging (AFI) system.

[0083] Another aspect of the invention provides a fiber optic imaging system for generating multipath image data. The system comprises: an optical coherence tomography (OCT) system comprising an OCT fiber optically connected to an interferometric detector; a multi-clad fiber (MCF) for receiving sampled light that has interacted with a sample and propagating the sampled light as a fundamental MCF mode and one or more higher order MCF modes; an optical joint for coupling the sampled light from the MCF into the OCT fiber, the optical joint providing an offset between the MCF and the OCT fiber to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber where the at least some light energy propagates as OCT return light.

[0084] The imaging system may comprise any of the features, combinations of features and/or sub-combinations of features discussed above or elsewhere herein.

[0085] Another aspect of the invention provides apparatus having any new and inventive feature, combination of features, or sub-combination of features described herein.

[0086] Another aspect of the invention provides methods having any new and inventive steps, acts, combination of steps and/or acts or sub-combination of steps and/or acts described herein.

[0087] It is emphasized that the invention relates to all combinations of the above features, even if these are recited in different claims.

[0088] Further aspects and example embodiments are illustrated in the accompanying drawings and/or described in the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0089] The accompanying drawings illustrate non-limiting example embodiments of the invention.

[0090] FIG. 1 is a schematic block diagram showing an fiber optic imaging system comprising an optical coherence tomography (OCT) system and a multi-clad fiber (MCF) according to an example embodiment.

[0091] FIG. 2A is a schematic diagram showing an optical joint having a spatial offset between two fibers which may be used in the imaging system of FIG. 1 according to an example embodiment.

[0092] FIG. 2B is a schematic diagram showing an optical joint having an angular offset between two fibers which may be used in the imaging system of FIG. 1 according to an example embodiment.

[0093] FIG. 3 is a schematic block diagram showing a fiber optic imaging system comprising multiple imaging types (one of which is an OCT system) and a multimode fiber (e.g. multi-clad fiber (MCF)) according to another example embodiment. FIG. 3A is a schematic block diagram of an optical joint of the FIG. 3 imaging system according to a particular embodiment.

[0094] FIGS. 4A and 4B are plots showing geometric and refractive properties of single-mode fiber (SMF) and double-clad fiber (DCF) respectively.

[0095] FIG. 5A is an enlarged view of an end of the sample arm of the FIG. 3 imaging system according to an example embodiment.

[0096] FIG. 5B is a perspective view of a visualization of OCT image data depicting the coordinates used to describe OCT image data according to an example embodiment.

[0097] FIG. 6 is a flowchart of a method for generating multipath image data (e.g. diagnostic image data) using an imaging system having a sample arm comprising a MCF (e.g. one of the imaging systems of FIG. 1 or FIG. 3) according to an example embodiment.

[0098] FIGS. 7A-D are schematic diagrams showing propagation of incident light, sampled light and OCT return light in a fiber optic imaging system according to an example embodiment.

[0099] FIG. 8A is a plan view of a so-called longitudinal section (A-line axis by pullback axis) of OCT image data corresponding to one azimuthal value according to an example implementation.

[0100] FIG. 8B is a plan view of an en face projection of fundamental mode data of OCT image data according to an example implementation.

[0101] FIG. 8C is a plan view of an en face projection of higher order mode data of OCT image data according to an example implementation.

[0102] FIG. 8D is a plan view of an en face multipath contrast image according to an example embodiment.

[0103] FIGS. 9A-E (collectively, FIG. 9) are depictions of a set of multi-imaging types experimental data obtained from in vivo imaging of a human lung.

[0104] FIGS. 10A-E (collectively, FIG. 10) are depictions of another set of multi-imaging types experimental data obtained from ex vivo imaging of a fallopian tube specimen.

[0105] FIGS. 11A-D (collectively, FIG. 11) are depictions of another set experimental data obtained from an ex vivo fallopian tube specimen which shows how multiple components of higher order mode image data may be extracted from the OCT image data.

DETAILED DESCRIPTION

[0106] Throughout the following description, specific details are set forth in order to provide a more thorough understanding of the invention. However, the invention may be practiced without these particulars. In other instances, well known elements have not been shown or described in detail to avoid unnecessarily obscuring the invention. Accordingly, the specification and drawings are to be regarded in an illustrative, rather than a restrictive sense.

[0107] Aspects of the invention provides systems and methods for generating multipath image data (e.g. diagnostic endoscopic image data) based on fundamental mode data and higher order mode data extracted from OCT image data.

[0108] One aspect of the invention provides a fiber optic imaging system for generating multipath image data (e.g. diagnostic endoscopic image data). The system comprises: an optical coherence tomography (OCT) system having an OCT fiber optically connected to an interferometric detector; a multi-clad fiber (MCF) for receiving sampled light that has interacted with a sample and propagating the sampled light as a fundamental MCF mode and one or more higher order MCF modes; and an optical joint for coupling the sampled light from the MCF into the OCT fiber. The optical joint is configured to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber where the at least some light energy propagates as OCT return light. The cross-coupling of light energy from one or more higher order MCF modes into the fundamental mode of the OCT fiber may be enabled by spatial and/or angular offset between the MCF and the OCT fiber. For example, such an offset may be between the optical axes of the MCF and the OCT fiber, where these optical axes represent the directions of light propagation.

[0109] The system may comprise a controller connected to the interferometric detector to receive, from the interferometric detector, OCT image data based on the OCT return light. The controller may be configured to: extract fundamental mode image data and higher order mode image data from the OCT image data, wherein the extraction is based on A-line coordinates of the OCT image data; and generate the multipath image data based on both of the fundamental mode image data and the higher order mode image data.

[0110] Another aspect of the invention provides a method for generating multipath image data (e.g. diagnostic image data) using the system(s) described herein. The method comprises: coupling the sampled light from a MCF into an optical coherence tomography (OCT) fiber, wherein coupling the sampled light comprises coupling at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber, where the at least some light energy propagates as OCT return light; generating OCT image data based on the OCT return light; extracting fundamental mode image data and higher order mode image data from the OCT image data, wherein the extracting is based on A-line coordinates of the OCT image data; and generating the multipath image data based on both of the fundamental mode image data and the higher order mode image data.

[0111] Another aspect of the invention provides achieving multi-type fiber optic imaging system (e.g. an endoscopic imaging system) comprising two or more imaging types, for example, OCT and AFL.

[0112] FIG. 1 is a schematic block diagram of a fiber optic (e.g. endoscopic) imaging system **100** according to an exemplary embodiment. System **100** comprises: an optical coherence tomography (OCT) system **102**; sampling optics **104**; a multi-clad fiber (MCF) **112**; and an optical joint **106** for coupling light between MCF **112** and OCT system **102**. System **100** may be used to generate multipath image data **119** from sample **111** (e.g. diagnostic image data from a body of a patient). For example, sampling optics **104** of system **100** may deliver incident light **121A** to a sample **111** (e.g. the body of the patient) and MCF **112** may collect reflected/backscattered light (referred to herein as sampled light) **121B** which has interacted with sample **111**.

[0113] System **100** may comprise any suitable OCT system. System **100** of the illustrated embodiment comprises at least one light source(s) **122**, typically a laser, and most typically a broadband source laser. In some embodiments, light source **122** comprises a swept-source laser. Light source **122** may emit light in any suitable wavelength range. Light as used herein should be understood to comprise electromagnetic radiation of any suitable wavelength. By way of non-limiting example, in some embodiments, light source **122** emits an infrared light. In a non-limiting example embodiment, light source **122** emits light in a wavelength range of about 1260 nm to about 1360 nm. System **100** of the illustrated embodiment also comprises an interferometric detector **126** for generating OCT image data **113** (which may be based on interferometric data (not explicitly shown) which is in turn based on interferometric optical path length differences) as is known to those skilled in the art. In some embodiments, interferometric data generated by interferometric detector **126** may be processed (e.g. by controller **120** and/or some other suitably configured processor (e.g. within OCT system **102**)) to generate OCT image data **113**.

[0114] System **100** further comprises an OCT fiber **116** which is used to couple incident light from light source **122** into sampling optics **104** via optical joint **106** and MCF **112**. OCT fiber **116** may comprise any suitable optical fiber. In some embodiments, OCT fiber **116** comprises a single-mode fiber (SMF). In some embodiments, OCT fiber **116** comprises a double-clad fiber (DCF). In some embodiments, OCT fiber **116** comprises a multi-clad fiber (MCF). In some embodiments, OCT fiber **116** comprises a few mode fiber (FMF).

[0115] Incident light from light source **122** propagates through OCT fiber **116** and is coupled into MCF **112** by optical joint **106**. In some embodiments, optical joint **106** comprises a fiber-optic rotary joint (FORJ). Any suitable FORJ may be used in optical joint **106**. MCF **112** is capable of propagating light in multiple modes. Incident light originating from light source **122** propagates through MCF **112** before being directed by sampling optics **104** to impinge on sample **111** as incident light **121A**.

[0116] In some embodiments, system **100** comprises a catheter (not expressly shown) in which MCF **112** may be embedded which extends into the body of a patient (sample **111**) for in vivo imaging. In some embodiments, optical joint **106** comprises a rotator configured to rotate MCF **112** to scan in azimuthal directions about a pullback axis **415** (described below in relation to FIG. 5B) which may be generally co-linear or parallel with axis **112A** of MCF **112** (also described below in relation to FIGS. 2A and 2B). Incident light propagating in MCF **112** and incident light **121A** that interacts with sample **111** (and incident light propagating in MCF **112**) may comprise a plurality of components; specifically, this incident light may comprise fundamental mode incident light (which may propagate primarily in the core of MCF **112**) and higher order mode incident light (which may propagate primarily in the cladding of MCF **112**, but which may additionally or alternatively comprise core mode(s)). These two components of incident light **121A** (and incident light propagating in MCF **112**) may arise because of optical joint **106**, which may cross-couple some light energy from the fundamental

mode of incident light from OCT fiber **116** to higher order mode incident light in MCF **112** (e.g. due to an offset between OCT fiber **116** and MCF **112** as explained in more detail below). This is not necessary, however. There may be a phase delay (positive or negative) between the two components of incident light in MCF **112** due to the different rates of propagation of the two components in MCF **112**.

[0117] In some embodiments, MCF **112** is capable of receiving high numerical aperture (NA) light—i.e. high NA light may be coupled into MCF **112** for propagation (e.g. multimode propagation) therein. For example, in some embodiments, MCF **112** is capable of receiving light with NA values in the range of up to about 0.2, 0.3 and/or higher. In some embodiments, high NA light received in MCF **112** is propagated in MCF **112** in multiple modes which include a fundamental mode which propagates primarily in a core (not explicitly enumerated) of MCF **112** and one or more higher order mode(s) which propagate primarily in one or more claddings of MCF **112**.

[0118] Incident light **121A** from MCF **112** is directed by sampling optics **104** to impinge onto, and interact with, sample **111**. MCF **112** receives reflected/backscattered light **121B** from sample **111** via sampling optics **104**. This reflected/backscattered light that has interacted with sample **111** may be referred to herein as sampled light. The sampled light **121B** received by MCF **112** may also comprise high NA light. The high NA sampled light received by MCF **112** may be propagated back in multiple modes by MCF **112** toward optical joint **106** and OCT system **102**. The multiple modes of sampled light propagating in MCF **112** may comprise a fundamental mode (referred to herein as a fundamental MCF mode) and one or more higher order modes (referred to herein a higher order MCF mode(s)). As with the incident light propagating in MCF **112**, the higher order MCF mode(s) of sampled light may have a plurality of components which include higher order MCF mode sampled light energy corresponding to fundamental mode incident light that has interacted with sample **111** and higher order MCF mode sampled light energy corresponding to higher order mode incident light that has interacted with sample **111**. As is known in the art, MCF **112** has the property that there is relative phase delay between the fundamental MCF mode and each of the higher order MCF mode(s) of sampled light propagating in MCF **112**—i.e. sampled light in different modes propagates at different speeds along MCF **112**. A total amount of phase delay between MCF modes of sampled light in MCF **112** will also depend on the length of MCF **112** in sample arm **104**. It will be appreciated that where the component of higher order MCF mode sampled light energy corresponds to higher order mode incident light, it will have twice the phase delay (positive or negative relative to the fundamental MCF mode sampled light) of the component of higher order MCF mode sampled light energy corresponding to fundamental mode incident light, since the former propagates in higher order modes of MCF **112** in both directions, whereas the latter propagates in the fundamental mode of MCF **112** in the forward direction (toward sample **111**) and in the higher order modes of MCF **112** in the reverse direction (away from sample **111**).

[0119] The high NA sampled light may comprise sampled light backscattered from sample **111** at relatively low angles which propagates primarily in the fundamental MCF mode and primarily in the core of MCF **112** and sampled light backscattered from sample **111** at relatively large angles which propagates primarily in the higher order MCF mode(s) and primarily in the one or more cladding(s) of MCF **112**.

[0120] Upon propagating back to optical joint **106** in MCF **112**, sampled light propagating in MCF **112** is coupled back into OCT fiber **116** by optical joint **106**. Joint **106** is configured to couple at least some of the light energy from both the fundamental MCF mode and the higher order MCF mode(s) into a fundamental mode of OCT fiber **116**, where the at least some light energy propagates (back toward interferometric detector **118**). The at least some light energy from both the fundamental MCF mode and the higher order MCF mode(s) that is coupled into the fundamental mode of OCT fiber **116** and that propagates back toward interferometric detector **118** may be referred to herein as OCT return light.

[0121] The coupling of at least some light energy from the higher order MCF mode(s) into the core of OCT fiber **116** may be referred to herein as cross-coupling. The light cross-coupled from the higher order MCF mode(s) into the core of OCT fiber **116** as OCT return light may comprise some of both components of the higher order MCF mode(s)—i.e. higher order MCF mode sampled light energy corresponding to higher order mode incident light and higher order MCF mode sampled light energy corresponding to fundamental mode incident light. Optical joint **106** may be configured in any suitable manner to cross-couple at least some light energy from higher order MCF mode(s) from MCF **112** into the fundamental mode of OCT fiber **116**. In some embodiments, optical joint **106** is provided by spatially and/or angularly offsetting OCT fiber **116** from MCF **112** to enable the cross-coupling.

[0122] FIG. 2A is a schematic view of optical joint **106** according to an example embodiment, where OCT fiber **116** is spatially offset from MCF **112**. More specifically, in the FIG. 2A embodiment, OCT axis **116A** of OCT fiber **116** is spatially offset from MCF axis **112A** of MCF **112** by a spatial offset **125**. FIG. 2B is a

schematic view of optical joint **106** according to another example embodiment, where OCT fiber **116** is angularly offset from MCF **112**. More specifically, in the FIG. 2B embodiment, OCT axis **116A** of OCT fiber **116** is angularly offset from MCF axis **112A** of MCF **112** by an angular offset **127**. As discussed above, optical joint **106** may comprise a fiber-optic rotary joint (FORJ). Details of the rotational aspects of optical joint **106** are omitted from FIGS. 2A and 2B.

[0123] In some embodiments, OCT fiber **116** is offset by a combination of spatial and angular offset with respect to MCF **112**. In some embodiments, other techniques may be used to effect the coupling of at least some of the light energy from the fundamental MCF mode from MCF **112** into the fundamental mode of OCT fiber **116** and cross-coupling at least some light energy from the one or more higher order MCF mode(s) from MCF **112** into the fundamental mode of OCT fiber **116**. By way of non-limiting example, such techniques may include: introducing one or more optical elements (e.g. one or more annuli, lenses, mirrors, mediums that cause refraction and/or the like) between MCF **112** and OCT fiber **116** to induce spatial/angular offsets or to otherwise affect the quality or shape of the sampled light emerging from MCF **112** before it enters OCT fiber **116**; twisting and/or bending one or both of MCF **112** and OCT fiber **116**; applying a mode scrambler to MCF **112** to scramble the higher order MCF mode(s) into the fundamental mode of OCT fiber **116**; inducing imperfections in one or both of MCF **112** and OCT fiber **116** (e.g. by pulling, heating, cleaving MCF **112** and/or OCT fiber **116** at an angle, and/or the like); and/or the like.

[0124] Returning to FIG. 1, OCT return light (i.e. having at least some light energy coupled from the fundamental MCF mode and the one or more higher order MCF mode(s) in MCF **112** into the fundamental mode of OCT fiber **116** by joint **106**) propagates back to OCT system **102**, where OCT system **102** is configured to use the OCT return light to generate OCT image data **113**. OCT image data **113** may comprise volumetric (e.g. tomographic) data related to sample **111**. In some embodiments, the OCT return light is directed to interferometric detector **126** which is configured to generate corresponding OCT image data **113** in accordance with any suitable OCT technique(s) known to those skilled in the art or subsequently developed in the art. In some embodiments, interferometric detector **126** may first generate interferometric data (not expressly shown in FIG. 1) based on the OCT return light and may comprise a suitably configured processor of any suitable type for processing the interferometric data to generate OCT image data **113** according to techniques known to those skilled in the art.

[0125] In some embodiments, interferometric detector **126** may generate interferometric data (not expressly shown in FIG. 1) and a controller **120** of imaging system **100** (which may be, in whole or in part, external to OCT system **102** or integrated, in whole or in part, within OCT system **102**) may be used to generate OCT image data **113** based on the interferometric data. Controller **120** should be understood to receive OCT image data **113** whether controller **120** is involved in generating OCT image data **113** or receives OCT image data from some separate system (e.g. OCT system **102**). In some embodiments, controller **120** comprises one or more stand-alone data processors and/or a stand-alone computer. In some embodiments, controller **120** comprises one or more data processors that are integrated into imaging system **100**. In general, however, controller **120** may comprise any suitably configured data processor as described elsewhere herein.

[0126] Whether OCT image data **113** is output from interferometric detector **126** and provided to controller **120**, or controller **120** is involved in the generation of OCT image data **113**, controller **120** receives OCT image data **113**. FIG. 8A shows a so-called longitudinal section comprising a pullback direction by A-line slice **800A** of exemplary OCT image data **113** corresponding to one azimuthal (ϕ) coordinate (i.e. one azimuthal value (ϕ) about pullback axis **415**). Because of the rotation of MCF **112** about its MCF axis **112A** (which may be generally coincident or generally parallel with) pullback axis **415**, OCT image data **113** comprises a plurality of sections similar to section **800A** (each having a different azimuthal coordinate (ϕ)).

[0127] It can be observed from the particular example of FIG. 8A, that there are two regions (region B and region C) of bright OCT image data **113** in slice **800A** and that these regions are separated from one another in the A-line direction. That is, bright region B generally has higher A-line coordinates than bright region C (In FIG. 8A, the origin of the A-line axis is at the top of the image and the A-line coordinate value increases towards the bottom of the page (as indicated by the arrow of A-line axis) in accordance with customary practice in the OCT field.). Bright regions B and C in OCT image data **113** are created by the different phase delays (introduced in MCF **112**) between the fundamental MCF mode light and the higher order MCF mode light. As discussed above, both the fundamental MCF mode and the higher order MCF mode(s) have some light energy coupled into the fundamental mode of OCT fiber **116** as OCT return light. Region B in the FIG. 8A slice **800A** of OCT image data **113** may correspond generally to OCT image data **113** associated with the fundamental MCF mode that is coupled into the fundamental mode of OCT fiber **116** and region C in the FIG. 8A slice **800A** of OCT image data **113** may correspond to generally to OCT image data **113** associated with the

higher order MCF mode(s) that is cross-coupled into the fundamental mode of OCT fiber **116**. The difference in the A-line coordinates of regions B and C may be “tuned” by controlling the total phase delay between the fundamental MCF mode light and the higher order MCF mode light in MCF **112** (e.g. by controlling the length and/or other propagation parameter(s) of MCF **112**). In some instances, region C may appear to translate with respect to region B when a sample arm position of system **100** is adjusted or if the spatial interferogram is sampled at a different frequency. Nonetheless, region C will generally be present in the reconstructed A-line due to aliasing. While not shown in the exemplary data of FIG. **8A**, in circumstances where light cross-coupled from the higher order MCF mode(s) into the core of OCT fiber **116** as OCT return light comprises some of both components of the higher order MCF mode(s)—i.e. higher order MCF mode sampled light energy corresponding to higher order mode incident light and higher order MCF mode sampled light energy corresponding to fundamental mode incident light, the FIG. **8A** slice **800A** of OCT image data (and similar slices at other azimuthal coordinates (ϕ)) may exhibit a third region of image data in a range of A-line coordinates corresponding to twice the phase delay of region C. This third region of OCT image data (with the largest phase delay—not shown in FIG. **8A**) may comprise image data related to the component of higher order MCF mode sampled light energy corresponding to higher order mode incident light, whereas region C shown in FIG. **8A** may comprise image data related to the component of higher order MCF mode sampled light energy corresponding to fundamental mode incident light. This OCT image data related to the component of higher order MCF mode sampled light energy corresponding to higher order mode incident light may be referred to herein as additional higher order mode image data **123**. Additional higher order mode image data **123** may be extracted from OCT image data **113** in a manner similar to higher order mode image data **117** (e.g. based on A-line coordinate thresholds). Multipath image data **119** may be based on fundamental mode image data **115** and either or both of higher order mode image data **117** and additional higher order mode image data **123**.

[0128] As explained in more detail below, controller **120** processes OCT image data **113** to: extract fundamental mode image data **115** and higher order mode image data **117** from OCT image data **113**; and generate multipath image data **119** based on both fundamental mode image data **115** and higher order mode image data **117**.

[0129] Extracting fundamental mode image data **115** and higher order mode image data **117** from OCT image data **113** may be based on the A-line coordinates of OCT image data **113**. For example, referring to the FIG. **8A** pullback by A-line slice **800A** of OCT image data **113**, fundamental mode image data **115** may correspond to region B and may be extracted by setting lower and upper A-line coordinate thresholds **804**, **802**—that is fundamental mode image data **115** may correspond to OCT image data **113** between lower A-line coordinate **804** and upper A-line coordinate **802**. Similarly, extracting higher order mode image data **117** from OCT image data **113** may be based on lower and upper A-line coordinate thresholds **808**, **806**. A-line coordinate thresholds **802**, **804**, **806**, **808** are configurable parameters of imaging system **100** and may depend on optical path length (e.g. total phase delay) differences between the fundamental MCF mode light and the higher order MCF mode light introduced by MCF **112**.

[0130] The inventors have discovered that higher order mode image data **117** may be used in conjunction with fundamental mode image data **115** to generate useful image data referred to herein as multipath image data **119**. As explained in more detail below, in some embodiments, generating multipath image data **119** based on both fundamental mode image data **115** and higher order mode image data **117** comprises: generating en face projections corresponding to fundamental mode image data **115** and higher order mode image data **117**; and determining multipath image data **119** to be a ratio of the en face fundamental mode image data projection to the en face higher order mode image data projection. This is not necessary. In some embodiments, other techniques based on both fundamental mode image data **115** and higher order mode image data **117** may be used to generate multipath image data **119**.

[0131] FIG. **3** is a schematic block diagram of a fiber optic imaging system (e.g. an endoscopic system) **200** according to another exemplary embodiment. In many respects, imaging system **200** is similar to imaging system **100** described elsewhere herein and elements and features of imaging system **200** are described using reference numerals that are similar to the reference numerals used in connection with imaging system **100** except that reference numerals of imaging system **200** are preceded by the digit “2” rather than “1” as is the case for features of imaging system **100**. For brevity, this description focusses on features of imaging system **200** that are different from those of imaging system **100**, it being understood that, unless the context dictates otherwise, features of imaging system **200** are analogous to those of imaging system **100**. Imaging system **200** differs from imaging system **100** in that imaging system **200** incorporates two imaging types provided by OCT system **202** and second imaging type **240**.

[0132] In imaging system **200**, OCT imaging system **202** and sampling optics **204** are substantially similar to OCT imaging system **102** and sampling optics **104** of imaging system **100**. Imaging system **200** is also similar to imaging system **100** in the manner that: sampled light which has interacted with sample **111** propagates back through MCF **212** (as fundamental MCF mode and one or more higher order MCF mode(s), which may have two components corresponding to higher order mode incident light and corresponding to fundamental mode incident light) and through optical joint **206** which couples at least some of the light energy from both the fundamental MCF mode and the higher order MCF mode(s) into the fundamental mode of OCT fiber **216** (as OCT return light); and OCT return light propagates back to OCT system **202**, where interferometric detector **226** generates OCT image data **213** (based on the OCT return light) and controller **220** processes OCT image data **213** to extract fundamental mode image data **215** and higher order mode image data **217** and to generate multipath image data **219** based on both fundamental mode image data **215** and higher order mode image data **217**.

[0133] Imaging system **200** differs from imaging system **100** in that imaging system **200** comprises a second imaging type provided by second imaging system **240** which: directs second imaging light through second imaging type fiber(s) **242** and optical joint **206**, into sampling optics **204** (and MCF **212**); receives second imaging type return light via second imaging type fiber(s) **242**; and generates second imaging type image data (e.g. second imaging type diagnostic image data) **244** based on the second imaging type return light. More specifically, second imaging system **240** comprises a second light source (not expressly shown in FIG. 3) which directs second incident light through second imaging type fiber(s) **242**, whereupon the second incident light is coupled into sampling optics **204** (and MCF **212**) by optical joint **206**. Light propagates in sampling optics **204** (and MCF **212**) in a similar manner to the light of system **100**, impinges on (and interacts with) sample **111** and high NA sampled light is coupled back into MCF **212** as fundamental MCF mode and one or more higher MCF modes. At optical joint **206**, at least some of the light energy from both the fundamental MCF mode and the higher order MCF mode(s) is coupled into the fundamental mode of OCT fiber **216** (as OCT return light), which is used to generate multipath image data **219** in a manner similar to imaging system **100**. However, optical joint **206** of the FIG. 3 system **200** also splits the light from the two imaging types, so that at least some of the light energy from the higher order MCF mode(s) is coupled into second imaging type fiber(s) **242** as second imaging type return light.

[0134] Second imaging system **240** receives the second imaging type return light from second imaging type fiber(s) **242** and uses the second imaging type return light to generate second image data **244**. In some embodiments, second imaging system **240** may comprise a suitably configured processor of any suitable type for processing the second imaging type return light to generate second image data **244** according to techniques known to those skilled in the art. In some embodiments, controller **220** of imaging system **200** (which may be external to OCT system **202** and second imaging system **240**) may be used to generate second image data **244** based on data obtained by second imaging system **240** from the second imaging type return light. In some embodiments, controller **220** comprises a standalone computer. In some embodiments, controller **220** comprises one or more data processors that are integrated into imaging system **200**. In general, however, controller **220** may comprise any suitably configured data processor as described elsewhere herein.

[0135] In some non-limiting embodiments, second imaging system **240** comprises an autofluorescence imaging (AFI) system **240**. In some such embodiments, second imaging type fiber(s) **242** comprise a first (input) fiber **242A** which provides second imaging type (AFI) incident light to optical joint **206**, MCF **212** and sampling optics **204** and a second (return) fiber **242B** which receives second imaging type (AFI) return light from sampling optics **204**, MCF **212** and optical joint **206**. This is not necessary. In some embodiments, a single second imaging type fiber **242** carries both second imaging type incident light and second imaging type return light.

[0136] As alluded to above optical joint **206** of the illustrated FIG. 3 imaging system **200** is similar to optical joint **106** of the FIG. 1 imaging system **100** in that optical joint **206** couples at least some of the light energy from both the fundamental MCF mode and the higher order MCF mode(s) into the fundamental mode of OCT fiber **216** (as OCT return light), which is used to generate multipath image data **219** in a manner similar to imaging system **100**. However, optical joint **206** of the FIG. 3 imaging system **200** differs from optical joint **106** of the FIG. 1 imaging system in that optical joint **206** also splits the light from the two imaging types, so that at least some of the light energy from the higher order MCF mode(s) is coupled into second imaging type fiber(s) **242** as second imaging type return light.

[0137] FIG. 3A schematically illustrates an optical joint **206** according to a non-limiting example embodiment. In the FIG. 3A embodiment, second imaging system **240** comprises an AFI system **240**, where second imaging type fiber(s) **242** comprise a first (input) fiber **242A** which provides second imaging type (AFI) incident light

to optical joint **206** and a second (return) fiber **242B** which receives second imaging type (AFI) return light from optical joint **206**. Like optical joint **106** described above, optical joint **206** comprises a FORJ **250** which performs the function of coupling light into and out from MCF **212** and sampling optics **204**. FORJ **250** may perform this coupling between MCF **212** and another intermediate MCF **252** which carries both incident AFI and OCT light and AFI return and OCT return light. FORJ **250** may be configured (e.g. to provide spatial and/or angular offset between MCF **212** and intermediate MCF **252** and/or otherwise configured) in a manner similar to that of optical joint **106** described herein to couple at least some light energy from the fundamental MCF mode and the higher order MCF mode(s) from MCF **212** into the fundamental mode of intermediate MCF **252**. FORJ **250** may also be configured to couple at least some light energy from the higher order MCF mode(s) from MCF **212** into cladding layers of intermediate MCF **252** which support higher order mode propagation.

[0138] Apart from FORJ, optical joint **206** of the FIG. 3A embodiment also comprises a wavelength division multiplexer (WDM) which multiplexes and de-multiplexes light of different wavelengths and a multi-clad fiber coupler (MCFC) which couples light from an input fiber into a multi-clad fiber and back from the multi-clad fiber back into an output fiber. Optical joint **206** functions as follows.

[0139] Incident OCT from light source **222** (FIG. 3) of OCT system **202** arrives at WDM **254** on OCT fiber **216**. Similarly, incident AFI light from the light source of AFI system **240** arrives at WDM **254** on AFI input fiber **242A**. In currently preferred embodiments, OCT fiber **216** comprises a SMF. As discussed above, this is not necessary, and OCT fiber **216** may comprise a SMF, DCF, MCF or FMF. Similarly, AFI input fiber **242A** may comprise a SMF, DCF, MCF or FMF. However, the AFI light and OCT light may have different wavelengths so that light from both OCT fiber **216** and AFI input fiber **242A** may be multiplexed onto fiber **256** by WDM **254**. Fiber **256** may also comprise a SMF, DCF, MCF or FMF. The combined (multiplexed) AFI and OCT incident light propagates on fiber **256** to MCFC **258** which couples the combined incident light onto intermediate MCF **252**. MCFC may be configured couple the combined incident light into the core of intermediate MCF **252**. The combined incident light then propagates through intermediate MCF **252** to FORJ **250** where it is coupled into MCF **212** and sampling optics **204** as described elsewhere herein.

[0140] Light interacts with sample **111** as described elsewhere herein and returns to FORJ **250** as sample light propagating in MCF **212** as fundamental MCF mode and higher order MCF mode(s), which may have two components corresponding to higher order mode incident light and corresponding to fundamental mode incident light. FORJ **250** is configured to couple sample light into both the core and the cladding(s) of MCF **252**. FORJ **250** is configured to couple at least some light energy from the fundamental MCF mode and the higher order MCF mode(s) from MCF **212** into the fundamental mode of intermediate MCF **252**. FORJ **250** may also be configured to couple at least some light energy from the fundamental MCF mode and the higher order MCF mode(s) from MCF **212** into cladding layers of intermediate MCF **252** which support higher order mode propagation. Return light from intermediate MCF **252** reaches MCFC **258**, where the return light is split. MCFC is configured to couple light having wavelengths of both OCT light source **222** and AFI light source from the cladding layer(s) of intermediate MCF **252** primarily into AFI return fiber **242B** which may comprise a multi-mode fiber (MMF) or MCF and to couple light from the core of intermediate MCF **252** primarily into the core of fiber **256**. AFI return light on AFI return fiber **242B** is detected by an AFI detector (cladding light having wavelength of OCT light source **222** is not detected by AFI detector) and used by AFI system to generate AFI image data **244** as described elsewhere herein. Return light propagating on fiber **256** having wavelengths of both OCT light source **222** and AFI light source passes through WDM **254** and is coupled onto OCT fiber **216** as OCT return light, which is in turn used to generate multipath image data **219** as described elsewhere herein. Return light propagating on fiber **256** having wavelength of AFI light source is not detected by the OCT system (i.e. not interfering in a manner to be detectable by the OCT system).

[0141] In some embodiments, one or more of the MCFs described herein (e.g. MCF **112**, MCF **212**, intermediate MCF **252**) may comprise double-clad fibers (DCFs). FIGS. 4A and 4B are plots **300A** and **300B** showing geometric and refractive index properties of SMF and DCF respectively. As shown in FIG. 4A, a SMF has two discrete indices of refraction: a first index of refraction, $n_{\text{sub.core}}$, in its core between the fiber axis and a radial core dimension $r_{\text{sub.core}}$; and a second, lower, index of refraction, $n_{\text{sub.clad}}$, in the radial annular cladding region between the radial core dimension $r_{\text{sub.core}}$ and a radius of the protective sheath of the fiber (not shown). In comparison, as shown in FIG. 4B, the DCF of the illustrated embodiment has three indices of refraction: a first index of refraction, $n_{\text{sub.core}}$, in its core between the fiber axis and a radial core dimension $r_{\text{sub.core}}$; a second, lower, index of refraction, $n_{\text{sub.clad1}}$, in an annular cladding region between the radial core dimension $r_{\text{sub.core}}$ and a second radial dimension $r_{\text{sub.clad1}}$; and a third, lower still, index of refraction, $n_{\text{sub.clad2}}$, in an annular cladding region between second radial dimension $r_{\text{sub.clad1}}$ and a radius

of the protective sheath of the fiber (not shown). In some embodiments, a fundamental mode of light energy propagates in the core of a DCF and higher order mode(s) of light energy propagate in the first cladding of the DCF.

[0142] In some embodiment, sampling optics (e.g. sampling optics **104** of system **100** and/or sampling optics **204** of system **200**) comprises an imaging end **400**. Imaging end **400** may comprise any suitable imaging end of any suitable endoscopic imaging system, for example. FIG. 5A shows an enlarged schematic cross-sectional view of an imaging end **400** of sampling optics **104** according to an example embodiment. Sampling optics **204** may comprise a similar imaging end **400**. FIG. 5B is a schematic perspective view of a visual representation of the coordinates used to describe OCT image data **113**, **213** generated by OCT systems **102**, **202** described herein according to an example embodiment.

[0143] In the example embodiment shown in FIG. 5A, imaging end **400** comprises a DCF region **402**, a multi-mode fiber (MMF) region **404**, a gradient index fiber (GRIN) region **406** and an angle-polished no-core fiber (NCF) region **408**. However, this is not necessary. Imaging end **400** depicted in the FIG. 5A embodiment represents one particular embodiment for an imaging end. In some embodiments, the imaging systems described herein may use other types of imaging ends which may comprise, by way of non-limiting example, any suitable imaging end of any suitable endoscopic system.

[0144] In operation, incident light **121A** (illustrated as solid straight arrow in FIG. 5A) is directed onto sample substantially along an A-line axis **411** (shown in FIG. 5B). To perform rotary scanning of a sample **111** (not shown in FIGS. 5A, 5B), sampling optics **104** and MCF **112** including imaging end **400** is rotated by optical joint **106** about MCF axis **112A** (pullback axis **415**) such that a region of incident light incident on sample **111** continuously traverses to different azimuthal (ϕ) coordinates **413** about MCF axis **112A** as MCF **112** is retracted from sample **111** in pullback direction **415** (which may be generally coincident with or generally parallel to MCF axis **112A**). The reflected/backscattered (sampled) light (illustrated as white arrows and dotted regions in FIG. 5A) is collected by MCF **112** and propagated through MCF **112** as multiple modes comprising a fundamental MCF mode described herein and one or more higher order MCF modes. Given the retracting rotary scanning, OCT image data **113**, **213** may have a helical scan pattern which may be approximated as a three-dimensional cylindrical shape as shown in FIG. 5B.

[0145] FIG. 6 is a flowchart of a method **600** for generating multipath image data (e.g. multipath image data **119**, **219**) using an imaging system having a MCF according to an example embodiment. FIG. 6 is described in part in conjunction with FIGS. 7A-7D (collectively, FIG. 7). FIG. 7 includes schematic diagrams of portions of an imaging system **700**, illustrating the propagation of light through imaging system **700** according to an example embodiment. Imaging system **700** may be similar to imaging systems **100**, **200** described above and reference numerals for features of imaging system **700** that are analogous to features of imaging systems **100**, **200** are similar to the reference numerals of imaging systems **100**, **200** except that reference numerals of imaging system **700** are preceded by the digit “7” rather than “1” (in the case of imaging system **100**) or “2” (in the case of the imaging system **200**). Features of imaging system **700** (e.g. particular fibers and/or other optical components) may be used in either of imaging systems **100**, **200**. For brevity, this description focusses on features of imaging system **700** that are different from those of imaging systems **100**, **200** it being understood that, unless the context dictates otherwise, features of imaging system **700** are analogous to those of imaging systems **100**, **200**. While the FIG. 6 method **600** is described in part in conjunction with imaging system **700**, it will be appreciated that method **600** could also be used with the other imaging systems (e.g. imaging systems **100**, **200**) described herein. Further, the explanation of the FIG. 6 method **600** in conjunction with the FIG. 7 optical system **700** I described with theoretical models of various features of optical system **700**. Neither method **600** nor optical system **700** are bound by any of these models.

[0146] Method **600** involves collecting image data from a sample using a MCF. The sample may be, but is not limited to, the body of a human or animal patient. The MCF may be any suitable MCF described herein (e.g. MCF **12**, **212**, **712**). In block **603**, incident light is directed, from the MCF, into the sample, where it interacts with the sample. The light propagation in block **603** is illustrated in MCF **712** shown in FIG. 7A. In some embodiments, MCF **712** comprises a DCF **712**, although this is not necessary and any suitable fiber supporting multimode propagation may be used. FIG. 7A also shows imaging system **700** incorporating an OCT system **702** comprising a Michelson interferometer **726**, although this also is not necessary and any suitable interferometry system may be used.

[0147] As shown in FIG. 7A, light source **722** emits incident light towards sample **111** along an optical axis **711**. In accordance with convention in the OCT field, optical axis **711** may refer to the direction that light is travelling at any point in system **700** and is often assigned the coordinate (z). It will be appreciated that optical axis **711** and/or its corresponding z coordinates may change orientations when the direction that light is

travelling in system **700** changes direction. In some embodiments, the incident light may be modelled as a polychromatic plane wave with electric field defined as:

$$[00001] E_i = s(k, \omega) e^{i(kz - \omega t)}$$

where $s(k, \omega)$ is the electric field amplitude as function of wavenumber $k=2\pi/\lambda$ and angular frequency $\omega=2\pi\nu$, λ is wavelength, ν is frequency and $i=\sqrt{-1}$.

[0148] Incident light from light source **722** impinges on beam splitter **742**, a reference reflector **748** and a detector **744**. Reference reflector **748** may be modelled as having an electric field reflectivity $r_{\text{sub.R}}$, and power reflectivity $R_{\text{sub.R}}=|r_{\text{sub.R}}|^2$. The reference path of interferometer **726** may be air and the distance from beam splitter **742** to reference reflector **748** may be referred to as $z_{\text{sub.R}}$.

[0149] Returning to FIG. **6**, in block **603**, the incident light is directed by sampling optics **704** from MCF **712** into sample **111**, where the incident light interacts with sample **111**. In block **605**, MCF **712** receives sampled light that has interacted with sample **111** and the sampled light propagates in multiple modes within MCF **712** (e.g. in a fundamental MCF mode and one or more higher order MCF modes). There is a relative phase delay between at least two of the multiple MCF modes propagating in MCF **712**. In some embodiments, the multiple modes of the reflected light correspond to the angles of reflection and/or backscattering of light that has interacted with sample **111**. For example, higher order MCF mode(s) may correspond primarily to higher angular backscattering and the fundamental MCF mode may correspond primarily to low angle backscattering. [0150] The process of block **605** is shown schematically in exemplary imaging system **700** of FIG. **7B**. Sample **111** under interrogation in FIG. **7** may be characterized by a depth- and angular-dependent reflectivity, $r_{\text{sub.S}}(x, y, z)$. As illustrated in FIG. **7B**, sample **111** may be modelled as having a series of N discrete angle-dependent reflections given by:

$$[00002] r_S(x, y, z) = \sum_{n=1}^N r_{S_n} A_n(x, y) \delta(z - z_{S_n})$$

with each reflection characterized by its electric field reflectivity $[r_{\text{sub.S.sub.1}}, r_{\text{sub.S.sub.2}}, \dots, r_{\text{sub.S.sub.n}}]$ or power reflectivity $R_{\text{sub.S.sub.n}}=|r_{\text{sub.S.sub.n}}|^2$, path length from beam splitter **742** $[z_{\text{sub.S.sub.1}}, z_{\text{sub.S.sub.2}}, \dots, z_{\text{sub.S.sub.n}}]$, and a complex envelope $[A_{\text{sub.1}}(x, y), A_{\text{sub.2}}(x, y), \dots, A_{\text{sub.N}}(x, y)]$ that encodes the angular distribution of the backscattered light. The Fourier transform of the complex envelope $\tilde{A}_{\text{sub.N}}(k_{\text{sub.x}}, k_{\text{sub.y}})$ describes the angular distribution of the backscattered light, where $k_{\text{sub.x}}=k\theta_{\text{sub.x}}$ and $k_{\text{sub.y}}=k\theta_{\text{sub.y}}$ are angular spatial frequencies along the x - and y -axes under the paraxial approximation, $\theta_{\text{sub.x}}$ is the angle between the wavevector and the yz -plane, and $\theta_{\text{sub.y}}$ (not shown in FIG. **7B**) is the angle between wavevector and the xz -plane.

[0151] When $A_{\text{sub.N}}(x, y)=1$, then $\tilde{A}_{\text{sub.N}}(k_{\text{sub.x}}, k_{\text{sub.y}})=\delta(k_{\text{sub.x}}, k_{\text{sub.y}})$, and the reflected light consists of backscattered light with scattering angle $\theta=\pm 180^\circ$. Similarly, a reflector with isotropic scattering has $A_{\text{sub.N}}(x, y)=\delta(x, y)$ and $\tilde{A}_{\text{sub.N}}(k_{\text{sub.x}}, k_{\text{sub.y}})=1$. A reflector that backscatters at an angle $\theta_{\text{sub.x}} < 180^\circ$ to an optical axis **711** will have $A_{\text{sub.N}}(x, y)=e^{i k \theta_{\text{sub.x}} x}$ and $\tilde{A}_{\text{sub.N}}(k_{\text{sub.x}}, k_{\text{sub.y}})=\delta(k_{\text{sub.x}} - k \theta_{\text{sub.x}}, k_{\text{sub.y}})$. The complex envelope $A_{\text{sub.N}}(x, y)$ corresponding to an arbitrary scattering phase function $p_{\text{sub.n}}(\theta)$ can be determined by first translating the coordinate axis by 180° to bring backscattered light to the origin then taking the Fourier transformation of the resulting phase function.

[0152] In the FIG. **7** embodiment, incident light is propagated in the fundamental mode of MCF **712** and projected into sample **111** by sampling optics **704** (shown in FIGS. **7A-C**). Azimuthally symmetric (the azimuthal coordinate denoted as ϕ in FIG. **7B**) sampled light backscattered along optical axis **711** is coupled into the fundamental mode of MCF **712** (fundamental MCF mode or, in the case of DCF **712**, fundamental DCF mode **712**). Sampled light backscattered at a non-zero angle θ to optical axis **711** may only be partially coupled into the fundamental mode, with some fraction of the optical power coupled into one or more higher order modes of MCF **712** (higher order MCF mode(s) or, in the case of DCF **712**, higher order DCF mode(s)).

[0153] The power fraction coupled into a particular mode depends on its spatial overlap with the incident electric field $A_{\text{sub.N}}(x, y)$. Using an overlap integral, the power coupled from discrete reflection n into linearly polarized mode $LP_{\text{sub.lm}}$ of MCF **712** may be modelled as:

$$[00003] \quad n, l, m = \frac{\int \int |A_N(x, y)|^2 dx dy \int \int |\tilde{A}_N(x, y)|^2 dx dy}{\int \int |\tilde{A}_N(x, y)|^2 dx dy \int \int |\tilde{A}_{l,m}(x, y)|^2 dx dy} \quad (1)$$

where $\psi_{\text{sub.l,m}}$ is the transverse mode profile of $LP_{\text{sub.lm}}$, l and m are the azimuthal and radial mode orders, and the integration is over the end face of MCF **712**.

[0154] Equivalently, the coupling coefficient can also be calculated in the spatial frequency domain:

$$[00004] \quad n, l, m = \frac{\int \int |\tilde{A}_N(k_x, k_y)|^2 dk_x dk_y \int \int |\tilde{A}_{l,m}(k_x, k_y)|^2 dk_x dk_y}{\int \int |\tilde{A}_N(k_x, k_y)|^2 dk_x dk_y \int \int |\tilde{A}_{l,m}(k_x, k_y)|^2 dk_x dk_y} \quad (2)$$

where $\{\tilde{\psi}\}_{\text{sub.l},m(k_{\text{sub.x}}, k_{\text{sub.y}})}$ is the Fourier transform of the transverse mode profile and the integration is over the angles supported by the aperture of sampling optics **704**.

[0155] Ideal backscatter ($\theta=\pm 180^\circ$, $\theta_{\text{sub.x}}=\theta_{\text{sub.y}}=0$) is coupled almost entirely into the fundamental mode and thus the coupling coefficient as defined above is close to one. Sampled light that is not as directly backscattered may partially couple into the fundamental mode and one or more higher order modes as characterized by $\alpha_{\text{sub.n},l,m}$. This angle-dependent mode coupling provides a means to differentiate scatterers via scattering angle distribution and may allow elucidation of the scattering angle distribution of the sampled light backscattered from a reflector.

[0156] Higher order modes with a non-zero azimuthal mode order (e.g. LP_{sub.11} in a step-index fiber) are sensitive to non-orthogonal incident light ($\theta_{\text{sub.x}}, \theta_{\text{sub.y}} > 0$) only if the reflector is displaced some small distance ($\delta_{\text{sub.x}}, \delta_{\text{sub.y}}$) from the optical axis **711**. Consequently, if such a reflector is on axis ($\delta_{\text{sub.x}}=\delta_{\text{sub.y}}=0$), the overlap integral given by equations. (1) and (2) is zero. In a scanning system, this means maximum sensitivity to a reflector with angle-dependent backscatter will occur when the reflector is slightly offset from optical axis **711**. To accommodate for this offset required for sensitivity, the mean coupling coefficient may be used:

$$[00005] \quad \tilde{\alpha}_{n,l,m} = \frac{\int \int \tilde{A}_N(k_x, k_y) \tilde{A}_{l,m}^*(k_x, k_y) dk_x dk_y}{\int \int \tilde{A}_N(k_x, k_y) dk_x dk_y \int \int \tilde{A}_{l,m}^*(k_x, k_y) dk_x dk_y} \quad \text{B}$$

as a surrogate for the ideal coupling coefficient defined above. The absolute square now operates directly on the Fourier transform of the complex envelope and transverse profile rather than the integral. This is equivalent to integrating over the offset-dependent coupling coefficient $\tilde{\alpha}_{\text{sub.n},l,m}(\delta_{\text{sub.x}}, \delta_{\text{sub.y}})$ and effectively removes the offset dependency from the coupling coefficients.

[0157] Referring to FIG. 7C, an angle-dependent reflection from each discrete reflector is decomposed by MCF **712** into a set of LP_{sub.lm} MCF modes **721** and **723-1** to **723-4** (or in the case of DCF **712**, DCF modes) which propagate down MCF **712**. Each MCF mode (e.g. fundamental MCF mode **721** and higher order MCF modes **723-1** to **723-4** shown in FIG. 7C) propagates with a unique effective refractive index and phase velocity resulting in relative phase delay between the different MCF modes. Higher order MCF modes **723-1** to **723-4** with effective refractive indices less than the fundamental MCF mode **721** will propagate faster and appear closer to the zero-reference delay in a processed OCT image, permitting modes which correspond to higher-angle backscatter to be identified given sufficient phase delay.

[0158] Returning to FIG. 6, in block **607**, the sampled light travelling in the MCF (as fundamental MCF mode (e.g. mode **721**) and one or more higher order MCF modes (e.g. modes **723-1** to **723-4**)) is directed to an optical joint **706**, which is configured to couple at least some light energy from both the fundamental MCF mode **721** and the one or more higher order MCF modes **723-1** to **723-4** from MCF **712** into a fundamental mode **725** of a fiber **716**, where the at least some light energy propagates as OCT return light. In some embodiments, fiber **716** is similar to OCT fiber **116** in that it is coupled directly to the OCT system. In some embodiments, fiber **716** is similar to intermediate fiber **252** that is coupled to a fiber coupler in the optical joint and indirectly coupled to the OCT system. Optical joint **706** is configured in any suitable manner to cross-couple at least some light energy from higher order MCF mode(s) **723-1** to **723-4** from MCF **712** into the fundamental mode **725** of fiber **716**. This block **607** process is shown schematically in FIG. 7D, where optical joint **706** comprises a FORJ.

[0159] Referring to FIG. 7D, optical joint **706** is configured to couple at least some light energy from the now-dispersed (because of phase delay) higher order MCF mode(s) **723-1** to **723-4** back into the fundamental mode **725** of fiber **716** (as OCT return light) which is directed to interferometer **726** to interfere with light from reference reflector **748**. Optical joint **706** may be configured to provide a slight offset ($\delta_{\text{sub.x}}, \delta_{\text{sub.y}}$) between the axes **712A**, **716A** of MCF **712** and fiber **716**. The overlap integral:

$$[00006] \quad \alpha_{l,m} = \frac{\int \int \alpha_{0,1}(x, y) \alpha_{l,m}^*(x \pm x, y \pm y) dx dy}{\int \int \alpha_{0,1}(x, y) dx dy \int \int \alpha_{l,m}^*(\pm x, \pm y) dx dy} \quad (4)$$

may be used to characterize the power coupling from LP_{sub.lm} to LP_{sub.01}, where the spatial-domain integration is over the end face of fiber **716**.

[0160] In block **609**, the OCT return light propagating in fiber **716** is directed to interferometer **726** to thereby generate OCT image data **713**.

[0161] Referring to FIG. 7, the electric field incident on beam splitter **742** and returning from reference reflector **748** is

$$[00007] E_R = \frac{E_i}{\sqrt{2}} r_R e^{i2kz_R},$$

while that the electric field returning from fiber **716** is:

$$[00008] E_S = \frac{E_i}{\sqrt{2}} \cdot \text{Math.}_{n=1}^N \cdot \text{Math.}_{l=0}^L \cdot \text{Math.}_{m=1}^M \sqrt{R_{S_n}} \sqrt{\frac{n, l, m}{l, m}} e^{ik(2z_{S_n} + n_{l,m} L_{DCF})} \quad (5)$$

[0162] The electric field at beam splitter **742** is the sum of all modes and all discrete reflectors. The factor of 2 in the first term of the exponential accounts for the roundtrip path length to each reflection. The second term, $\Delta n_{\text{sub.l,mL.sub.DCF}}$, is non-zero for higher-order modes, reducing the effective path length of the reflection and translating it in the negative z direction. The effective refractive index difference, $\Delta n_{\text{sub.l,m=n.sub.l,m-n.sub.0,1}}$, is definitionally negative for higher order modes, where $n_{\text{sub.l,m}}$ is the effective refractive index of mode (l, m). The physical length of MCF **712** $L_{\text{sub.DCF}}$ is multiplied by the mode-dependent refractive index difference to calculate the phase offset resulting in a translation of the higher order modes reflection in the negative z direction.

[0163] Detector **744** generates a photocurrent

$$[00009] I_D = \frac{1}{\tau} \cdot \text{Math.} \cdot \text{Math.} E_R + E_S \cdot \text{Math.}^2 \cdot \text{Math.}$$

proportional to the square of the sums of the reference and sample electric fields, where ρ is the responsivity of detector **744**, and the angled brackets represent time averaging. The spectral interferogram contains DC, cross-correlation, and auto-correlation terms. The cross-correlation terms,

$$[00010] I_{\text{OCT}}(k) = \quad (6)$$

$$\frac{1}{\tau} S(k) \left\{ \begin{aligned} & \text{Math.}_{n=1}^N \sqrt{R_R R_{S_n}} \sqrt{\frac{n, 0, 1}{0, 1}} \cos[2k(z_R - z_{S_n})] + \\ & \text{Math.}_{n=1}^N \cdot \text{Math.}_{l=1}^L \cdot \text{Math.}_{m=1}^M \sqrt{R_R R_{S_n}} \sqrt{\frac{n, l, m}{l, m}} \cos[k(2(z_R - z_{S_n}) + n_{l,m} L_{DCF})] \end{aligned} \right\}$$

encode the position, reflectivity, and angular distribution of the reflectors, where $S(k) = |s(k, \omega)|^2$ is the power spectrum of the source. Note that in equation (6), the term owing to the fundamental mode ($l=0, m=1$) has been written explicitly to distinguish it from the terms responsible for the higher order modes. An inverse Fourier transform can then be applied to recover the A-line profile:

[00011]

$$i_{\text{OCT}}(z) = \frac{1}{\tau} \gamma(z) \cdot \text{Math.} \left\{ \begin{aligned} & \text{Math.}_{n=1}^N \sqrt{R_R R_{S_n}} \sqrt{\frac{n, 0, 1}{0, 1}} [z \pm 2(z_R - z_{S_n})] + \\ & \text{Math.}_{n=1}^N \cdot \text{Math.}_{l=1}^L \cdot \text{Math.}_{m=1}^M \sqrt{R_R R_{S_n}} \sqrt{\frac{n, l, m}{l, m}} [z \pm 2(z_R - z_{S_n}) + n_{l,m} L_{DCF}] \end{aligned} \right\} \quad (7)$$

where $\gamma(z)$ is the coherence (or point spread) function, $\cdot$ is the convolution operator, and δ is the Dirac delta function.

[0164] Cross-coupling by optical joint **706** of at least some light energy from one or more higher order MCF modes **723-1** to **723-4** in MCF **712** into the fundamental mode **725** of fiber **716** as OCT return light is highest when there is a large spatial overlap in the geometric area of the core of the respective fibers **712** and **716**. In some embodiments, azimuthal modes of order $l=1$ are cross-coupled sufficiently (i.e. high $\beta_{\text{sub.l,m}}$) into fundamental mode **725** of fiber **716** for detection through interference with the reference. As described elsewhere herein, this results in higher order mode image data **717** which is consistently composed of the same higher order modes, and therefore appears in the same location relative to the fundamental mode image data **715** within a given A-line, which permits the extraction of the fundamental mode image data **715** and higher order mode image data **717** from the OCT image data **713** based on A-line coordinates.

[0165] Returning to FIG. **6**, in block **611**, fundamental mode image data **715** and higher order mode image data **717** are extracted from OCT image data **713**. The extraction of fundamental mode image data and higher order mode image data is based on A-line coordinates of OCT image data **713** as described elsewhere herein. The block **611** extraction may be performed by a controller (see controllers **120, 220** in FIGS. **1** and **2**).

[0166] In block **613**, the controller generates multipath image data **719** based on both of fundamental mode image data **715** and higher order mode image data **717**. In some embodiments, generating multipath image data **719** based on both of fundamental mode image data **715** and higher order mode image data **717** comprises obtaining a ratio between fundamental mode image data **715** and higher order mode image data **717** (i.e. contrast between fundamental mode image data **715** and higher order mode image data **717**). In some embodiments, fundamental mode image data **715** comprises a fundamental mode image (FI) and higher order mode image data **717** comprises a higher order mode image (HOI).

[0167] In a non-limiting example embodiment, multipath contrast (MC) is obtained to leverage the differential coupling of sampled light into the FI and the HOI. MC is defined as the ratio between the optical powers coupled into the FI versus the HOI, which can be formulated from equation (7) as follows:

$$[00012] \text{ MC} = \frac{\text{Math}_{n=1}^N \cdot \sqrt{R_{S_n}} \cdot \sqrt{n,0,1} \cdot 0,1}{\text{Math}_{n=1}^N \cdot \text{Math}_{l=1}^L \cdot \text{Math}_{m=1}^M \cdot \sqrt{R_{S_n}} \cdot \sqrt{n,l,m} \cdot l,m} \quad (8)$$

[0168] The MC is a function of the cross-coupling efficiencies $\beta_{\text{sub},l,m}$ through a discontinuity (optical joint **106, 206, 706**) and the FI and HOI coupling efficiencies $\alpha_{\text{sub},n,l,m}$. Notably, the ratio-metric nature of equation (8) removes any dependency on the reflectivity of the reference arm, thus limiting the variance in MC response from changes in the reference arm position or focus. Assuming the modal cross-coupling efficiencies of optical joint **106, 206, 706** remain constant between images, MC is proportional to the ratio of a backscattered electric field sampled at low angles (by the LP.sub.01 mode) over the field sampled at higher angles (HOMs of the form LP.sub.11-like).

[0169] FIG. **8A** is a plan view of a so-called longitudinal section (i.e. a pullback by A-line section) **800A** of example OCT image data **113, 213, 713** corresponding to one azimuthal (ϕ) coordinate (i.e. one azimuthal value (ϕ) about pullback axis **415**). It will be appreciated that OCT image data **113, 213, 713** comprises a plurality of sections similar to section **800A** (each having a different azimuthal coordinate (ϕ)). FIG. **8B** is a plan view of an “en face” projection **800B** of fundamental mode image data **115, 215, 715** of the exemplary FIG. **8A** OCT image data **113, 213, 713** according to an example embodiment. FIG. **8C** is a plan view of an en face projection **800C** of higher order mode image data **117, 217, 717** of the exemplary FIG. **8A** OCT image data **113, 213, 713** according to an example embodiment. FIG. **8D** is a plan view of an en face projection **800D** of a multipath contrast image based on the FIG. **8B** fundamental mode image data **115, 215, 715** and the FIG. **8C** higher order mode image data **117, 217, 717** according to an example embodiment.

[0170] FIGS. **8A-D** (collectively, FIG. **8**) are depictions of exemplary data obtained from imaging a human fingertip. Pullback by A-line longitudinal section **800A** (FIG. **8A**) is shown in a plane defined by an A-line axis (e.g. A-line **411** shown in FIG. **5B**) corresponding to one azimuthal (ϕ) coordinate (i.e. one azimuthal value (ϕ) about pullback axis **415**) and a pullback axis **415** (see FIG. **5B**), which is the direction in which the MCF **112, 212, 712** is pulled back from sample **111**. Because of the rotation of MCF **112, 212, 712** about its own axis (which may be coincident or parallel with pullback axis **415**), OCT image data **113** comprises a plurality of sections similar to section **800A** (each having a different azimuthal coordinate (ϕ)). Therefore, pullback by A-line section **800A** may be thought of as one pullback by radial (or pullback by A-line) slice (corresponding to one azimuthal value (ϕ)) of example OCT image data **113, 213, 713**, it being understood that there are a plurality of similar pullback by A-line slices (one for each azimuthal value (ϕ)) in OCT image data **113, 213, 713**.

[0171] It can be observed from FIG. **8A**, that there are two regions (region B and region C) of bright OCT image data **113, 213, 713** in slice **800A** and that these regions are separated from one another in the A-line direction. That is, bright region B generally has higher A-line coordinates than bright region C. The spatial offset between bright regions B and C in OCT image data **113, 213, 713** is created by the different phase delays (introduced in MCF **112, 212, 712**) between the fundamental MCF mode light and the higher order MCF mode light. The unique transverse profile of each mode provides angular sensitivity (i.e. the modes are sensitive to slightly different incident EM fields backscattered from the sample and therefore captures scattering differently). As discussed above, both the fundamental MCF mode and the higher order MCF mode(s) have some light energy coupled into the fundamental mode of OCT fiber **116, 216, 716** as OCT return light. Region B in the FIG. **8A** slice **800A** of OCT image data **113, 213, 713** may correspond generally to OCT image data **113, 213, 713** associated with the fundamental MCF mode that is coupled into the fundamental mode of OCT fiber **116, 216, 716** and region C in the FIG. **8A** slice **800A** of OCT image data **113, 213, 713** may correspond to generally to OCT image data **113, 213, 713** associated with the higher order MCF mode(s) that is cross-coupled into the fundamental mode of OCT fiber **116, 216, 716**. The difference in the A-line coordinates of regions B and C may be “tuned” by controlling the total phase delay between the fundamental MCF mode light and the higher order MCF mode light in MCF **112, 212, 712** (e.g. by controlling the length and/or other propagation parameter(s) of MCF **112, 212, 712**). In some instances, region C may appear to translate with respect to Region B when a sample arm position of system **100** is adjusted or if the spatial interferogram is sampled at a different frequency. Nonetheless, region C will generally be present in the reconstructed A-line due to aliasing. While not shown in the exemplary data of FIG. **8A**, in circumstances where light cross-coupled from the higher order MCF mode(s) into the core of OCT fiber **116, 216, 716** as OCT return light comprises some of both components of the higher order MCF mode(s)—i.e. higher order MCF mode sampled light energy corresponding to higher order mode incident light and higher order MCF mode sampled light energy corresponding to fundamental mode incident light, the FIG. **8A** slice **800A** of OCT image data (and similar slices at other azimuthal coordinates (ϕ)) may exhibit a third region of image data in a range of A-line

coordinates corresponding to twice the phase delay of region C. This third region of OCT image data (with the largest phase delay—not shown in FIG. 8A) may comprise image data related to the component of higher order MCF mode sampled light energy corresponding to higher order mode incident light, whereas region C shown in FIG. 8A may comprise image data related to the component of higher order MCF mode sampled light energy corresponding to fundamental mode incident light. This OCT image data related to the component of higher order MCF mode sampled light energy corresponding to higher order mode incident light may be referred to herein as additional higher order mode image data **123, 223, 723**. Additional higher order mode image data **123, 223, 723** may be extracted from OCT image data **113, 213, 713** in a manner similar to higher order mode image data **117, 217, 717** (e.g. based on A-line coordinate thresholds). Multipath image data **119, 219, 719** may be based on fundamental mode image data **115, 215, 715** and either or both of higher order mode image data **117, 217, 717** and additional higher order mode image data **123, 223, 723**.

[0172] As explained above, OCT image data **113, 213, 713** is processed to: extract fundamental mode image data **115, 215, 715** and higher order mode image data **117, 217, 717** from OCT image data **113, 213, 713** (see block **611** of method **600** in FIG. 6); and generate multipath image data **119, 219, 719** based on both fundamental mode image data **115, 215, 715** and higher order mode image data **117, 217, 717** (see block **613** of method **600** in FIG. 6).

[0173] Extracting fundamental mode image data **115, 215, 715** and higher order mode image data **117, 217, 717** from OCT image data **113, 213, 713** may be based on the A-line coordinates of OCT image data **113, 213, 713**. For example, referring to the FIG. 8A pullback by A-line slice **800A** of OCT image data **113, 213, 713**, fundamental mode image data **115, 215, 715** may correspond to region B and may be extracted by setting lower and upper A-line coordinate thresholds **804, 802**—that is fundamental mode image data **115, 215, 715** may correspond to OCT image data **113, 213, 713** between lower A-line coordinate **804** and upper A-line coordinate **802**. Similarly, extracting higher order mode image data **117, 217, 717** from OCT image data **113, 213, 713** may be based on lower and upper A-line coordinate thresholds **808, 806**. A-line coordinate thresholds **802, 804, 806, 808** are configurable parameters of imaging system **100** and may depend on optical path length (e.g. total phase delay) differences between the fundamental MCF mode light and the higher order MCF mode light introduced by MCF **112, 212, 712**.

[0174] The inventors have discovered that higher order mode image data **117, 217, 717** may be used in conjunction with fundamental mode image data **115, 215, 715** to generate useful image data referred to herein as multipath image data **119, 219, 719**. In some embodiments, generating multipath image data **119, 219, 719** based on both based on both fundamental mode image data **115, 215, 715** and higher order mode image data **117, 217, 717** comprises: generating en face projections corresponding to fundamental mode image data **115, 215, 715** and higher order mode image data **117, 217, 717**; and determining multipath image data **119, 219, 719** to be a ratio of the en face fundamental mode image data projection to the en face higher order mode image data projection. This is not necessary. In some embodiments, other techniques based on both fundamental mode image data **115, 215, 715** and higher order mode image data **117, 217, 717** may be used to generate multipath image data **119, 219, 719**.

[0175] FIG. 8B shows an exemplary en face projection **800B** of fundamental mode image data **115, 215, 715** extracted from the exemplary OCT image data **113, 213, 713** shown in FIG. 8A (and the additional OCT image data **113, 213, 713** corresponding to other azimuthal (ϕ) coordinates) on the basis of A-line coordinates. FIG. 8A shows an exemplary region B of longitudinal section **800A** from which a portion of en face projection **800B** can be extracted. Region B of A-line by pullback longitudinal section **800A** can be seen to be a region of relatively higher A-line coordinates (when compared to region C of A-line by pullback longitudinal section **800A**) between a lower A-line coordinate threshold **804** and an upper A-line coordinate threshold **802**. As discussed above, the A-line by pullback longitudinal section **800A** of FIG. 8A represents a single azimuthal value (ϕ) in OCT image data **113, 213, 713**. The en face projection **800B** of FIG. 8B may be obtained by: for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in OCT image data **113, 213, 713** (i.e. for each azimuthal, pullback coordinate pair in OCT image data **113, 213, 713**), summing or averaging or otherwise aggregating or processing values along the A-line direction between lower A-line coordinate threshold **804** and upper A-line coordinate threshold **802** (e.g. in region B shown in FIG. 8A) to obtain a value for an en face pixel corresponding to the current azimuthal, pullback coordinate pair. Since the A-line by pullback longitudinal section **800A** shown in FIG. 8A represents only one azimuthal value (ϕ) of OCT image data **113, 213, 713**, the data of FIG. 8A only creates one horizontal line of pixels in the FIG. 8B en face image **800B**. En face image **800B** is a projection in the pullback and azimuthal coordinate (ϕ) axes.

[0176] Exemplary en face projection **800C** of higher order mode image data **117, 217, 717** is extracted from the exemplary OCT image data **113, 213, 713** on the basis of A-line coordinates in a manner similar to en face

projection **800B** of fundamental mode image data **115, 215, 715** described above. FIG. **8A** shows an exemplary region C of longitudinal section **800A** from which a portion of en face projection **800C** can be extracted. Region C of A-line by pullback longitudinal section **800A** can be seen to be a region of relatively lower A-line coordinates (when compared to region B of A-line by pullback longitudinal section **800A**) between a lower A-line coordinate threshold **808** and an upper A-line coordinate threshold **806**. As discussed above, the A-line by longitudinal section **800A** of FIG. **8A** represents a single azimuthal value (ϕ) in OCT image data **113, 213, 713**. The en face projection **800C** of FIG. **8C** may be obtained by: for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in OCT image data **113, 213, 713** (i.e. for each azimuthal, pullback coordinate pair in OCT image data **113, 213, 713**), summing or averaging values along the A-line direction between lower A-line coordinate threshold **808** and upper A-line coordinate threshold **806** (e.g. in region C shown in FIG. **8A**) to obtain a value for an en face pixel corresponding to the current azimuthal, pullback coordinate pair. Since the A-line by pullback longitudinal section **800A** shown in FIG. **8A** represents only one azimuthal value (ϕ) of OCT image data **113, 213, 713**, the data of FIG. **8A** only creates one horizontal line of pixels in the FIG. **8C** en face image **800C**. En face image **800C** is a projection in the pullback and azimuthal coordinate (ϕ) axes.

[0177] Comparing the exemplary fundamental mode en face projection **800B** (FIG. **8B**) and exemplary the higher order mode en face projection **800C** (FIG. **8C**), bright ridges **811** of en face projection **800B** correspond to dark creases **821** in en face projection **800C** and dark creases **813** in en face projection **800B** correspond to bright ridges **823** in en face projection **800C**.

[0178] En face images (e.g. en face projections **800B, 800C**) may be used to generate multipath image data **119, 219, 719** using any suitable techniques. In one exemplary technique, a ratio of fundamental mode and higher order mode en face projections (e.g. en face projections **800B, 800C**) may be used to generate multipath image data **119, 219, 719**. FIG. **8D** shows exemplary multipath image data **119, 219, 719** shown as a multipath en face projection **800D** obtained by pixel-wise division between en face projection **800B** and en face projection **800C**. As can be observed in image **800D**, the brighter ridges **831** and darker creases **833** are more strongly emphasized in multipath en face projection **800D** by using both fundamental mode image data **115, 215, 715** and higher order mode image data **117, 217, 717** (e.g. by using exemplary en face projections **800B, 800C**). The low-angle scattering from ridges **831** result in a relative increase in coupling into the fundamental mode, compared to the higher angled scattering from the creases which correspond more to the higher order modes. The angle at which the sample is imaged impacts the multipath contrast response.

[0179] FIGS. **9A-E** (collectively, FIG. **9**) are depictions of experimental data obtained from imaging a human lung sample exhibiting an adenocarcinoma using a multi-imaging type imaging system of the type disclosed herein (e.g. system **200** comprising an OCT imaging system **202** and a second imaging type system **240** comprising an AFI imaging system **240**). The left-hand side of FIGS. **9A-E** is a distal end (i.e. deeper into sub-segmental airway) and the right-hand side of FIGS. **9A-E** is proximal (i.e. closer to a bronchoscope) along the pullback direction. FIG. **9A** depicts AFI image data **944** (i.e. a specific example of second type image data **244**), FIG. **9B** depicts OCT image data **913** (e.g. similar to OCT image data **113, 213, 713**), FIGS. **9C** and **9D** respectively depict en face projections of fundamental mode image data **915** (e.g. similar to fundamental mode image data **115, 215, 715**) and higher order mode image data **917** (e.g. similar to higher order image mode data **117, 217, 717**) and FIG. **9D** depicts an en face projection of multipath image data **919** (e.g. similar to multipath image data **119, 219, 719**).

[0180] Pullback by azimuthal section **900A** of AFI image data **944** (FIG. **9A**) is shown in a plane defined by azimuthal (ϕ) axis and the pullback axis). Pullback by A-line section **900B** (FIG. **9B**) of OCT image data **913** is shown in a plane defined by an A-line axis (e.g. A-line **411** shown in FIG. **5B**) corresponding to one azimuthal (ϕ) coordinate (i.e. one azimuthal value (ϕ) about pullback axis) and a pullback axis (e.g. axis **415** shown in FIG. **5B**). Section **900B** shown in FIG. **9B** represents data from line B in FIG. **9A** (and corresponding lines for other azimuthal coordinates (ϕ)). Line B in FIG. **9A** bisects darkened autofluorescence region **945** shown in FIG. **9A**, therefore visualizing a thickened epithelial layer. FIG. **9C** shows an en face projection **900C** of fundamental mode image data **915** extracted from OCT image data **913** shown in FIG. **9B** (and the additional OCT image data **913** corresponding to other azimuthal (ϕ) coordinates) on the basis of A-line coordinates. FIG. **9B** shows an exemplary region C of longitudinal section **900B** from which a portion of en face projection **900C** can be extracted. FIG. **9D** shows an en face projection **900D** of higher order mode image data **917** extracted from OCT image data **913** shown in FIG. **9B** (and the additional OCT image data **913** corresponding to other azimuthal (ϕ) coordinates) on the basis of A-line coordinates. FIG. **9B** also shows an exemplary region D of longitudinal section **900B** from which a portion of en face projection **900D** of FIG. **9D** can be extracted. The en face projections of FIGS. **9C** and **9D** may be obtained from OCT image data **913** in a manner similar to the en face projections described elsewhere herein with reference to FIG. **8**. A ratio of fundamental mode and

higher order mode en face projections (e.g. en face projections **900C**, **900D**) may be used to generate multipath image data **919** shown as a multipath en face projection **900E** in FIG. **9E**.

[0181] In the FIG. **9A** AFI image data **1044**, it can be observed that a region **945** of AFI image data **944** is noticeably relatively darker than other regions in section AFI image data **944**. However, there may be many causes for a darkened region in AFI image data **944**. Therefore, it is difficult to draw a definitive diagnostic conclusion about the darkened region **945** on the basis of AFI image data **944** alone. In both en face projections **900C** and **900D**, region **945** is not easily distinguishable from surrounding regions. However, in multipath image data **919** of FIG. **9E**, there is a visible difference between higher contrast region **949** and surrounding regions, suggesting that there is a decrease in backscattering angle over region **949** compared to the surrounding tissue. The higher contrast region **949** within region **945** of multipath image data **919** corroborates the darkened region **945** observed in AFI image data **944**.

[0182] FIGS. **10A-E** (collectively, FIG. **10**) are depictions of another set of experimental data obtained from ex vivo imaging of a fallopian tube specimen using a multi-imaging type imaging system of the type disclosed herein (e.g. system **200** comprising an OCT imaging system **202** and a second imaging type system **240** comprising an AFI imaging system **240**). The left-hand side of FIGS. **10A-E** is the isthmus (i.e. uterine end) and the right-hand side of FIGS. **10A-E** is the fimbriae (i.e. ovarian) along the pullback direction. FIG. **10A** depicts AFI image data **1044** (i.e. a specific example of second type image data **244**), FIG. **10B** depicts OCT image data **1013** (e.g. similar to OCT image data **113**, **213**, **713**), FIGS. **10C** and **10D** respectively depict en face projections of fundamental mode image data **1015** (e.g. similar to fundamental mode image data **115**, **215**, **715**) and higher order mode image data **1017** (e.g. similar to higher order image mode data **117**, **217**, **717**) and FIG. **10D** depicts an en face projection of multipath image data **1019** (e.g. similar to multipath image data **119**, **219**, **719**).

[0183] Pullback by azimuthal section **1000A** of AFI image data **1044** (FIG. **10A**) is shown in a plane defined by azimuthal (ϕ) axis and the pullback axis). Pullback by A-line section **1000B** (FIG. **10B**) of OCT image data **1013** is shown in a plane defined by an A-line axis corresponding to one azimuthal (ϕ) coordinate (i.e. one azimuthal value (ϕ) about pullback axis) and a pullback axis (e.g. pullback axis **415** shown in FIG. **5B**). FIG. **10C** is an en face projection **1000C** of fundamental mode image data **1015** extracted from OCT image data **1013** shown in FIG. **10B** (and the additional OCT image data **1013** corresponding to other azimuthal (ϕ) coordinates) on the basis of A-line coordinates. FIG. **10B** shows an exemplary region C of longitudinal section **1000B** from which a portion of en face projection **1000C** can be extracted. FIG. **10D** shows an en face projection **1000D** of higher order mode image data **1017** extracted from OCT image data **1013** shown in FIG. **10B** (and the additional OCT image data **1013** corresponding to other azimuthal (ϕ) coordinates) on the basis of A-line coordinates. FIG. **10B** also shows an exemplary region D of longitudinal section **1000B** from which a portion of en face projection **1000D** of FIG. **10D** can be extracted. The en face projections of FIGS. **10C** and **10D** may be obtained from OCT image data **1013** in a manner similar to the en face projections described elsewhere herein with reference to FIG. **8**. A ratio of fundamental mode and higher order mode en face projections (e.g. en face projections **1000C**, **1000D**) may be used to generate multipath image data **1019** shown as a multipath en face projection **1000E** in FIG. **10E**.

[0184] In the FIG. **10A** AFI image data **1044**, it can be observed that region **1045** is darkened compared to the surrounding regions of section **1000A**. Darkened region **1045** may therefore be a potential indication of high grade serous carcinoma (HGSC). However, it can be seen that such feature is not clearly distinguishable in longitudinal section **1000B** of OCT image data **1013**, en face projection **1000C** of fundamental mode image data **1015** or en face projection **1000D** of higher order mode image data **1017** of FIGS. **9B-D** respectively. Nevertheless, it can be observed that in multipath image data **1019** shown in en face projection **900E** of FIG. **9E**, a darkening is observable in region **1045** compared to the surrounding regions of image data **1019**. Therefore, multipath image data **1019** may provide corroboration to AFI image data **1044** (e.g. identification of HGSC in fallopian tubes).

[0185] In some embodiments, sample **111**, **211**, **711** is the tissue of a human or animal subject and imaging systems **100**, **200**, **700** comprise endoscopic imaging systems for use in generating multipath image data **119**, **219**, **719** that is useful for diagnosis of various medical conditions. This is not necessary. While sample **111**, **211**, **711** may be tissue of a human or animal subject, sample **111**, **211**, **711** may comprise other suitable matter for which imaging is desired. While imaging systems **100**, **200**, **700** may comprise endoscopic imaging systems, imaging systems **100**, **200**, **700** may comprise fiber optic imaging systems **100**, **200**, **700** generally. While multipath image data **119**, **219**, **719** may be useful for diagnosis of various medical conditions, this is not necessary and multipath image data **119**, **219**, **719** may be used for any suitable purpose.

[0186] FIGS. **11A-D** (collectively, FIG. **11**) are depictions of another set experimental data obtained from an ex

vivo fallopian tube specimen which shows how multiple components of higher order mode image data (corresponding to higher order mode incident light and fundamental mode incident light) may be extracted from the OCT image data. In the FIG. 11 images of the fallopian tube, the left hand side is the isthmus (uterine end) and the right hand side is the fimbriated (ovarian) end. FIG. 11A shows an A-line by pullback longitudinal section of OCT image data 113, 213, 713 having three regions B, C and D separable by A-line coordinate thresholds. Region B shown in FIG. 11A represents fundamental mode image data 115, 215, 715 that corresponds to image data where sample 111 is illuminated by fundamental mode incident light which is coupled back into the fundamental mode of MCF 212. FIG. 11B is an en face image corresponding to region B in FIG. 11A. Region C shown in FIG. 11A represents higher order mode image data 117, 217, 717 that corresponds to image data where sample 111 is illuminated by fundamental mode incident light which is coupled back into the higher order mode(s) of MCF 212. FIG. 11C is an en face image corresponding to region C in FIG. 11A. Region D shown in FIG. 11A represents additional higher order mode image data 122, 222, 722 that corresponds to image data where sample 111 is illuminated by higher order mode incident light which is coupled back into the higher order mode(s) of MCF 212. FIG. 11C is an en face image corresponding to region C in FIG. 11A. Multipath image data 119 may be based on fundamental mode image data 115, 215, 715 and either or both of higher order mode image data 117, 217, 717 and additional higher order mode image data 122, 222, 722.

[0187] Where a component (e.g. a software module, processor, assembly, device, circuit, etc.) is referred to herein, unless otherwise indicated, reference to that component (including a reference to a “means”) should be interpreted as including as equivalents of that component any component which performs the function of the described component (i.e., that is functionally equivalent), including components which are not structurally equivalent to the disclosed structure which performs the function in the illustrated exemplary embodiments of the invention.

[0188] Embodiments of the invention may be implemented using specifically designed hardware, configurable hardware, programmable data processors configured by the provision of software (which may optionally comprise “firmware”) capable of executing on the data processors, special purpose computers or data processors that are specifically programmed, configured, or constructed to perform one or more steps in a method as explained in detail herein and/or combinations of two or more of these. Examples of specifically designed hardware are: logic circuits, application-specific integrated circuits (“ASICs”), large scale integrated circuits (“LSIs”), very large scale integrated circuits (“VLSIs”), and the like. Examples of configurable hardware are: one or more programmable logic devices such as programmable array logic (“PALs”), programmable logic arrays (“PLAs”), and field programmable gate arrays (“FPGAs”). Examples of programmable data processors are: microprocessors, digital signal processors (“DSPs”), embedded processors, graphics processors, graphics processing units (GPUs), math co-processors, general purpose computers, server computers, cloud computers, mainframe computers, computer workstations, and the like. For example, one or more data processors in a control circuit for a device may implement methods as described herein by executing software instructions in a program memory accessible to the processors.

[0189] Processing may be centralized or distributed. Where processing is distributed, information including software and/or data may be kept centrally or distributed. Such information may be exchanged between different functional units by way of a communications network, such as a Local Area Network (LAN), Wide Area Network (WAN), or the Internet, wired or wireless data links, electromagnetic signals, or other data communication channel.

[0190] Aspects of the invention may also be provided in the form of a program product. The program product may comprise any non-transitory medium which carries a set of computer-readable instructions which, when executed by a data processor, cause the data processor to execute a method of the invention. Program products according to the invention may be in any of a wide variety of forms. The program product may comprise, for example, non-transitory media such as magnetic data storage media including floppy diskettes, hard disk drives, optical data storage media including CD ROMs, DVDs, electronic data storage media including ROMs, flash RAM, EPROMs, hardwired or preprogrammed chips (e.g., EEPROM semiconductor chips), nanotechnology memory, or the like. The computer-readable signals on the program product may optionally be compressed or encrypted.

[0191] In some embodiments, the invention may be implemented in software. For greater clarity, “software” includes any instructions executed on a processor, and may include (but is not limited to) firmware, resident software, microcode, code for configuring a configurable logic circuit, applications, apps, and the like. Both processing hardware and software may be centralized or distributed (or a combination thereof), in whole or in part, as known to those skilled in the art. For example, software and other modules may be accessible via local

memory, via a network, via a browser or other application in a distributed computing context, or via other means suitable for the purposes described above.

[0192] Software and other modules may reside on servers, workstations, personal computers, tablet computers, and other devices suitable for the purposes described herein.

[0193] The invention comprises a number of non-limiting aspects. Non-limiting aspects of the invention comprise:

[0194] 1. A fiber optic imaging system for generating multipath image data, the system comprising: [0195] an optical coherence tomography (OCT) system comprising an OCT fiber optically connected to an interferometric detector; [0196] a multi-clad fiber (MCF) for receiving sampled light that has interacted with a sample and propagating the sampled light as a fundamental MCF mode and one or more higher order MCF modes; [0197] an optical joint for coupling the sampled light from the MCF into the OCT fiber, the optical joint configured to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber where the at least some light energy propagates as OCT return light.

[0198] 2. The imaging system of aspect 1 or any other aspect herein comprising: [0199] a controller connected to the interferometric detector to receive, from the interferometric detector, OCT image data based on the OCT return light, the controller configured to: [0200] extract fundamental mode image data and higher order mode image data from the OCT image data, wherein the extraction is based on A-line coordinates of the OCT image data; and [0201] generate multipath image data (e.g. diagnostic image data) based on both of the fundamental mode image data and the higher order mode image data.

[0202] 3. The imaging system of aspect 2 or any other aspect herein wherein the controller is configured to: [0203] extract the fundamental mode image data based on light energy from the fundamental MCF mode that is coupled into the fundamental mode of the OCT fiber; and [0204] extract the higher order mode image data based on light energy from the one or more higher order MCF modes that is coupled into the fundamental mode of the OCT fiber.

[0205] 4. The imaging system of any one of aspects 1 to 3 or any other aspect herein wherein the controller is configured to extract the fundamental mode image data and the higher order mode image data from the OCT image data using upper and lower A-line coordinate thresholds.

[0206] 5. The imaging system of any one of aspects 1 to 4 or any other aspect herein wherein the MCF introduces a phase delay between the fundamental MCF mode and the one or more higher order MCF modes.

[0207] 6. The imaging system of aspect 5 or any other aspect herein wherein the controller is configured to extract the fundamental mode image data and the higher order mode image data based on the phase delay between the fundamental MCF mode and the one or more higher order MCF modes.

[0208] 7. The imaging system of any one of aspects 5 to 6 or any other aspect herein wherein an amount of the phase delay depends on a length of the MCF and wherein the amount of the phase delay is sufficient to enable extraction of the fundamental mode image data and the higher order mode image data from the OCT image data based on the A-line coordinates of the OCT image data.

[0209] 8. The imaging system of any one of aspects 1 to 7 or any other aspect herein wherein the MCF comprises a MCF core and at least one light-transmitting MCF cladding and wherein the fundamental MCF mode propagates substantially in the MCF core and the one or more higher order MCF modes propagate substantially in the at least one MCF cladding.

[0210] 9. The imaging system of any one of aspects 1 to 8 or any other aspect herein wherein the fundamental MCF mode comprises light energy received from relatively low numerical aperture portions of the sampled light and the one or more higher order MCF modes comprise light energy received from relatively high numerical aperture portions of the sampled light.

[0211] 10. The imaging system of any one of aspects 1 to 9 or any other aspect herein wherein the MCF propagates incident light from the optical joint toward the sample in an incident direction opposite to a direction of propagation of the sampled light, at least a portion of the incident light interacting with the sample to become the sampled light.

[0212] 11. The imaging system of aspect 10 or any other aspect herein wherein the incident light comprises: fundamental mode incident light that propagates in the incident direction in a fundamental mode of the MCF; and higher order mode incident light that propagates in the incident direction in one or more higher order modes of the MCF.

[0213] 12. The imaging system of aspect 11 or any other aspect herein wherein the one or more higher order MCF modes of the sampled light comprise a first component corresponding to the fundamental mode incident light that has interacted with the sample and a second component corresponding to the higher order mode

incident light that has interacted with the sample.

[0214] 13. The imaging system of aspect 12 of any other aspect herein wherein the controller is configured to extract the higher order mode image data based on light energy from the first component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.

[0215] 14. The imaging system of aspect 13 or any other aspect herein wherein the controller is configured to extract additional higher order mode image data from the OCT image data based on light energy from the second component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.

[0216] 15. The imaging system of aspect 14 or any other aspect herein wherein the controller is configured to extract the additional higher order mode image data based on A-line coordinates of the OCT image data.

[0217] 16. The imaging system of any one of aspects 14 to 15 or any other aspect herein wherein the controller is configured to extract the fundamental mode image data, the higher order mode image data and the additional higher order mode image data based on differential phase delay between different propagation modes introduced in the MCF.

[0218] 17. The imaging system of any one or aspects 14 to 16 or any other aspect herein wherein the controller is configured to generate multipath image data (e.g. diagnostic image data) based at least in part on the additional higher order mode image data.

[0219] 18. The imaging system of any one of aspects 1 to 17 or any other aspect herein wherein the optical joint is configured to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into the fundamental of the OCT fiber by introducing an offset between the MCF and the OCT fiber.

[0220] 19. The imaging system of aspect 18 or any other aspect herein wherein the offset comprises a spatial offset between an axis of the MCF and an axis of the OCT fiber.

[0221] 20. The imaging system of any one of aspects 18 to 19 or any other aspect herein wherein the offset comprises an angular offset between an axis of the MCF and an axis of the OCT fiber.

[0222] 21. The imaging system of any one of aspects 18 to 20 or any other aspect herein wherein the optical joint comprises one or more optical components (e.g. lenses and/or mirrors) which introduce optical offset between the MCF and the OCT fiber.

[0223] 22. The imaging system of any one of aspects 1 to 21 or any other aspect herein wherein the optical joint comprises a fiber optic rotary joint (FORJ) configured to rotate the MCF about its axis relative to the sample.

[0224] 23. The imaging system of any one of aspects 1 to 22 or any other aspect herein wherein the fundamental mode image data and the higher order mode image data are volumetric data.

[0225] 24. The imaging system of any one of aspects 1 to 23 or any other aspect herein wherein the controller is configured to: [0226] process the fundamental mode image data to obtain two-dimensional processed fundamental mode image data; [0227] process the higher order mode image data to obtain two-dimensional processed higher order mode image data; and [0228] generate the multipath image data based on the two-dimensional processed fundamental mode image data and the two-dimensional processed higher order mode image data.

[0229] 25. The imaging system of aspect 24 or any other aspect herein wherein the controller is configured to: [0230] process the fundamental mode image data to obtain the two-dimensional processed fundamental mode image data as a fundamental mode en face image by repeating: for each for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in the fundamental mode image data, aggregating (e.g. adding or averaging) pixel values along the A-line direction between lower and upper fundamental mode A-line coordinate thresholds to obtain one corresponding pixel value of the two-dimensional processed fundamental mode image data; and [0231] process the higher order mode image data to obtain the two-dimensional processed higher order mode image data as a higher order mode en face image by repeating: for each for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in the higher order mode image data, aggregating (e.g. adding or averaging) pixel values along the A-line direction between lower and upper higher order mode A-line coordinate thresholds to obtain one corresponding pixel value of the two-dimensional processed higher order mode image data.

[0232] 26. The imaging system of aspect 25 or any other aspect herein wherein the controller is configured to generate the multipath image data based on a combination of the fundamental mode en face image and the higher order mode en face image.

[0233] 27. The imaging system of any one of aspects 25 to 26 or any other aspect herein wherein the controller is configured to generate the multipath image data based on a ratio of the fundamental mode en face image and

the higher order mode of an en face image.

[0234] 28. The imaging system of any one of aspects 1 to 27 or any other aspect herein wherein MCF comprises a double-cladded fiber (DCF).

[0235] 29. The imaging system of any one of aspects 1 to 28 or any other aspect herein comprising a second imaging type system, the second imaging type system configured to direct second incident light from the optical joint toward the sample by propagating the second incident light in the MCF, wherein at least a portion of the second incident light interacts with the sample to become second sampled light and wherein the second imaging type system is configured to collect the second sampled light which propagates back toward the optical joint in the MCF.

[0236] 30. The imaging system of aspect 29 or any other aspect herein comprising a first light source that generates first incident light, at least a portion of which interacts with the sample to become the sampled light and wherein the second imaging type system comprises a second light source for generating the second incident light.

[0237] 31. The imaging system of aspect 30 or any other aspect herein wherein a wavelength of the first incident light is different than a wavelength of the second incident light.

[0238] 32. The imaging system of aspect 3 or any other aspect herein wherein the optical joint comprises a wavelength division multiplexer configured to combine the first incident light and the second incident light to generate combined incident light.

[0239] 33. The imaging system of aspect 32 or any other aspect herein wherein the optical joint comprises a fiber optic rotary joint (FORJ) configured to rotate the MCF about its axis relative to the sample and for coupling the combined incident light into the MCF.

[0240] 34. The imaging system of aspect 33 or any other aspect herein wherein the optical joint comprises a multi-clad fiber coupler (MCFC) configured to separate the sampled light from the second sampled light.

[0241] 35. The imaging system of any one of aspects 29 to 34 wherein the second imaging type system comprises an optical detector configured to detect at least a portion of the second sampled light and wherein the controller is configured to generate a second type image based on the detected portion of the second sampled light.

[0242] 36. The imaging system of aspect 35 or any other aspect herein wherein the second type image comprises an en face image.

[0243] 37. The imaging system of any one of aspects 29 to 36 or any other aspect herein wherein the second imaging type system is an autofluorescence imaging (AFI) system.

[0244] 38. A method for generating multipath image data, the method comprising: [0245] providing a multi-clad fiber (MCF); [0246] receiving sampled light that has interacted with the sample and propagating the sampled light in the MCF as a fundamental MCF mode and one or more higher order MCF modes; [0247] coupling the sampled light from the MCF into an optical coherence tomography (OCT) fiber, wherein coupling the sampled light comprises coupling at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber, where the at least some light energy propagates as OCT return light; [0248] generating OCT image data based on the OCT return light; [0249] extracting fundamental mode image data and higher order mode image data from the OCT image data, wherein the extracting is based on A-line coordinates of the OCT image data; and [0250] generating the multipath image data based on both of the fundamental mode image data and the higher order mode image data.

[0251] 39. The method of aspect 38 or any other aspect herein wherein: [0252] the fundamental mode image data is based on light energy from the fundamental MCF mode that is coupled into the fundamental mode of the OCT fiber; and [0253] the higher order mode image data is based on light energy from the one or more higher order MCF modes that is coupled into the fundamental mode of the OCT fiber.

[0254] 40. The method of any one of aspects 38 to 39 or any other aspect herein wherein extracting the fundamental mode image data and higher order mode image data from the OCT image data comprises using upper and lower A-line thresholds.

[0255] 41. The method of any one of aspects 38 to 40 or any other aspect herein wherein the MCF introduces a phase delay between the fundamental MCF mode and the one or more higher order MCF modes.

[0256] 42. The method of aspect 41 or any other aspect herein wherein extracting the fundamental mode image data and higher order mode image data from the OCT image data comprises is based on the phase delay between the MCF mode and the one or more higher order MCF modes.

[0257] 43. The method of any one of aspects 41 to 42 or any other aspect herein wherein an amount of the phase delay depends on a length of the MCF and wherein the amount of the phase delay is sufficient to enable

extraction of the fundamental mode image data and the higher order mode image data from the OCT image data based on the A-line coordinates of the OCT image data.

[0258] 44. The method of any one of aspects 38 to 43 or any other aspect herein wherein the MCF comprises a MCF core and at least one light-transmitting MCF cladding and wherein the fundamental MCF mode propagates substantially in the MCF core and the one or more higher order MCF modes propagate substantially in the at least one MCF cladding.

[0259] 45. The method of any one of aspects 38 to 44 or any other aspect herein wherein the fundamental MCF mode comprises light energy received from relatively low numerical aperture portions of the sampled light and the one or more higher order MCF modes comprise light energy received from relatively high numerical aperture portions of the sampled light.

[0260] 46. The method of any one of aspects 38 to 45 or any other aspect herein wherein the MCF propagates incident light toward the sample in an incident direction opposite to a direction of propagation of the sampled light, at least a portion of the incident light interacting with the sample to become the sampled light.

[0261] 47. The method of aspect 46 or any other aspect herein wherein the incident light comprises: fundamental mode incident light that propagates in the incident direction in a fundamental mode of the MCF; and higher order mode incident light that propagates in the incident direction in one or more higher order modes of the MCF.

[0262] 48. The method of aspect 47 or any other aspect herein wherein the one or more higher order MCF modes of the sampled light comprise a first component corresponding to the fundamental mode incident light that has interacted with the sample and a second component corresponding to the higher order mode incident light that has interacted with the sample.

[0263] 49. The method of aspect 48 or any other aspect herein wherein extracting the higher order mode image data is based on light energy from the first component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.

[0264] 50. The method of aspect 49 or any other aspect herein comprising extracting additional higher order mode image data from the OCT image data based on light energy from the second component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.

[0265] 51. The method of aspect 50 or any other aspect herein comprising extracting the additional higher order mode image data based on A-line coordinates of the OCT image data.

[0266] 52. The method of any one of aspects 50 to 51 or any other aspect herein comprising extracting the fundamental mode image data, the higher order mode image data and the additional higher order mode image data based on differential phase delay between different propagation modes introduced in the MCF.

[0267] 53. The method of any one of aspects 50 to 52 or any other aspect herein wherein generating the multipath image data (e.g. diagnostic image data) is based at least in part on the additional higher order mode image data.

[0268] 54. The method of any one of aspects 38 to 53 or any other aspect herein wherein coupling at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into the fundamental of the OCT fiber comprises providing an offset between the MCF and the OCT fiber.

[0269] 55. The method of aspect 54 or any other aspect herein wherein the offset comprises a spatial offset between an axis of the MCF and an axis of the OCT fiber.

[0270] 56. The method of any one of aspects 54 to 55 or any other aspect herein wherein the offset comprises an angular offset between an axis of the MCF and an axis of the OCT fiber.

[0271] 57. The method of any one of aspects 54 to 56 or any other aspect herein wherein providing the offset comprises locating one or more optical components (e.g. lenses and/or mirrors) in an optical path between the MCF and the OCT fiber to thereby introduce optical offset between the MCF and the OCT fiber.

[0272] 58. The method of any one of aspects 38 to 57 or any other aspect herein wherein coupling at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into the fundamental of the OCT fiber comprises locating a fiber optic rotary joint (FORJ) configured to rotate the MCF about its axis relative to the sample in an optical path between the MCF and the OCT fiber.

[0273] 59. The method of any one of aspects 38 to 58 or any other aspect herein wherein the fundamental mode image data and the higher order mode image data are volumetric data.

[0274] 60. The method of any one of aspects 38 to 59 or any other aspect herein comprising: [0275] processing the fundamental mode image data to obtain two-dimensional processed fundamental mode image data; [0276] processing the higher order mode image data to obtain two-dimensional processed higher order mode image data; and [0277] generating the multipath image data based on the two-dimensional processed fundamental

mode image data and the two-dimensional processed higher order mode image data.

[0278] 61. The method of aspect 60 or any other aspect herein wherein: [0279] processing the fundamental mode image data to obtain the two-dimensional processed fundamental mode image data comprises processing the fundamental mode image data to obtain a fundamental mode en face image by repeating: for each for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in the fundamental mode image data, aggregating (e.g. adding or averaging) pixel values along the A-line direction between lower and upper fundamental mode A-line coordinate thresholds to obtain one corresponding pixel value of the two-dimensional processed fundamental mode image data; and [0280] processing the higher order mode image data to obtain the two-dimensional processed higher order mode image data comprises processing the higher order mode image data to obtain a higher order mode en face image by repeating: for each for each azimuthal coordinate (ϕ) and for each pullback axis coordinate in the higher order mode image data, aggregating (e.g. adding or averaging) pixel values along the A-line direction between lower and upper higher order mode A-line coordinate thresholds to obtain one corresponding pixel value of the two-dimensional processed higher order mode image data.

[0281] 62. The method of aspect 61 or any other aspect herein wherein generating the multipath image data comprises generating the multipath image data based on a combination of the fundamental mode en face image and the higher order mode en face image.

[0282] 63. The method of any one of aspects 61 to 62 or any other aspect herein wherein generating the multipath image data comprises generating the multipath image data based on a ratio of the fundamental mode en face image and the higher order mode en face image.

[0283] 64. The method of any one of aspects 38 to 63 or any other aspect herein wherein the MCF comprises a double-cladded fiber (DCF).

[0284] 65. The method of any one of aspects 38 to 64 or any other aspect herein comprising: [0285] providing a second imaging type system; [0286] propagating second incident light toward the sample by propagating the second incident light in the MCF; [0287] wherein at least a portion of the second incident light interacts with the sample to become second sampled light; and [0288] collecting the second sampled light which propagates back toward the second imaging type system in the MCF.

[0289] 66. The method of aspect 65 or any other aspect herein comprising: [0290] generating first incident light by a first light source, at least a portion of which interacts with the sample to become the sampled light; and [0291] generating the second incident light by a second light source.

[0292] 67. The method of aspect 66 or any other aspect herein wherein a wavelength of the first incident light is different than a wavelength of the second incident light.

[0293] 68. The method of aspect 67 or any other aspect herein comprising combining the first incident light and the second incident light to generate combined incident light.

[0294] 69. The method of aspect 68 or any other aspect herein comprising coupling the combined incident light into the MCF by a fiber optic rotary joint (FORJ).

[0295] 70. The method of aspect 69 or any other aspect herein comprising separating the sampled light from the second sampled light using a multi-clad fiber coupler (MCFC).

[0296] 71. The method of any one of aspects 65 to 70 or any other aspect herein comprising: [0297] detecting at least a portion of the second sampled light; and [0298] generating a second type image based on the detected portion of the second sampled light.

[0299] 72. The method of aspect 71 or any other aspect herein wherein the second type image comprises an en face image.

[0300] 73. The method of any one of aspects 65 to 72 or any other aspect herein wherein the second imaging type system is an autofluorescence imaging (AFI) system.

[0301] 74. A fiber optic imaging system for generating multipath image data, the system comprising: [0302] an optical coherence tomography (OCT) system comprising an OCT fiber optically connected to an interferometric detector; [0303] a multi-clad fiber (MCF) for receiving sampled light that has interacted with a sample and propagating the sampled light as a fundamental MCF mode and one or more higher order MCF modes; [0304] an optical joint for coupling the sampled light from the MCF into the OCT fiber, the optical joint providing an offset between the MCF and the OCT fiber to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber where the at least some light energy propagates as OCT return light.

[0305] 75. The imaging system of aspect 40 comprising any of the features, combinations of features and/or sub-combinations of features of any of aspects 1 to 37.

[0306] 76. Apparatus having any new and inventive feature, combination of features, or sub-combination of

features described herein.

[0307] 77. Methods having any new and inventive steps, acts, combination of steps and/or acts or sub-combination of steps and/or acts as described herein.

Interpretation of Terms

[0308] Unless the context clearly requires otherwise, throughout the description and the claims: [0309] “comprise”, “comprising”, and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to”; [0310] “connected”, “coupled”, or any variant thereof, means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof; [0311] “herein”, “above”, “below”, and words of similar import, when used to describe this specification, shall refer to this specification as a whole, and not to any particular portions of this specification; [0312] “or”, in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list; [0313] the singular forms “a”, “an”, and “the” also include the meaning of any appropriate plural forms. These terms (“a”, “an”, and “the”) mean one or more unless stated otherwise; [0314] “and/or” is used to indicate one or both stated cases may occur, for example A and/or B includes both (A and B) and (A or B); [0315] “approximately” when applied to a numerical value means the numerical value $\pm 10\%$; [0316] where a feature is described as being “optional” or “optionally” present or described as being present “in some embodiments” it is intended that the present disclosure encompasses embodiments where that feature is present and other embodiments where that feature is not necessarily present and other embodiments where that feature is excluded. Further, where any combination of features is described in this application this statement is intended to serve as antecedent basis for the use of exclusive terminology such as “solely,” “only” and the like in relation to the combination of features as well as the use of “negative” limitation(s) to exclude the presence of other features; and [0317] “first” and “second” are used for descriptive purposes and cannot be understood as indicating or implying relative importance or indicating the number of indicated technical features.

[0318] Words that indicate directions such as “vertical”, “transverse”, “horizontal”, “upward”, “downward”, “forward”, “backward”, “inward”, “outward”, “left”, “right”, “front”, “back”, “top”, “bottom”, “below”, “above”, “under”, and the like, used in this description and any accompanying claims (where present), depend on the specific orientation of the apparatus described and illustrated. The subject matter described herein may assume various alternative orientations. Accordingly, these directional terms are not strictly defined and should not be interpreted narrowly.

[0319] Where a range for a value is stated, the stated range includes all sub-ranges of the range. It is intended that the statement of a range supports the value being at an endpoint of the range as well as at any intervening value to the tenth of the unit of the lower limit of the range, as well as any subrange or sets of sub ranges of the range unless the context clearly dictates otherwise or any portion(s) of the stated range is specifically excluded. Where the stated range includes one or both endpoints of the range, ranges excluding either or both of those included endpoints are also included in the invention.

[0320] Certain numerical values described herein are preceded by “about”. In this context, “about” provides literal support for the exact numerical value that it precedes, the exact numerical value $\pm 5\%$, as well as all other numerical values that are near to or approximately equal to that numerical value. Unless otherwise indicated a particular numerical value is included in “about” a specifically recited numerical value where the particular numerical value provides the substantial equivalent of the specifically recited numerical value in the context in which the specifically recited numerical value is presented. For example, a statement that something has the numerical value of “about 10” is to be interpreted as: the set of statements: [0321] in some embodiments the numerical value is 10; [0322] in some embodiments the numerical value is in the range of 9.5 to 10.5; and if from the context the person of ordinary skill in the art would understand that values within a certain range are substantially equivalent to 10 because the values with the range would be understood to provide substantially the same result as the value 10 then “about 10” also includes: [0323] in some embodiments the numerical value is in the range of C to D where C and D are respectively lower and upper endpoints of the range that encompasses all of those values that provide a substantial equivalent to the value 10

[0324] Specific examples of systems, methods and apparatus have been described herein for purposes of illustration. These are only examples. The technology provided herein can be applied to systems other than the example systems described above. Many alterations, modifications, additions, omissions, and permutations are possible within the practice of this invention. This invention includes variations on described embodiments that would be apparent to the skilled addressee, including variations obtained by: replacing features, elements and/or acts with equivalent features, elements and/or acts; mixing and matching of features, elements and/or

acts from different embodiments; combining features, elements and/or acts from embodiments as described herein with features, elements and/or acts of other technology; and/or omitting combining features, elements and/or acts from described embodiments.

[0325] As will be apparent to those of skill in the art upon reading this disclosure, each of the individual embodiments described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any other described embodiment(s) without departing from the scope of the present invention.

[0326] Any aspects described above in reference to apparatus may also apply to methods and vice versa.

[0327] Any recited method can be carried out in the order of events recited or in any other order which is logically possible. For example, while processes or blocks are presented in a given order, alternative examples may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or subcombinations. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed in parallel, simultaneously or at different times.

[0328] Various features are described herein as being present in “some embodiments”. Such features are not mandatory and may not be present in all embodiments. Embodiments of the invention may include zero, any one or any combination of two or more of such features. All possible combinations of such features are contemplated by this disclosure even where such features are shown in different drawings and/or described in different sections or paragraphs. This is limited only to the extent that certain ones of such features are incompatible with other ones of such features in the sense that it would be impossible for a person of ordinary skill in the art to construct a practical embodiment that combines such incompatible features. Consequently, the description that “some embodiments” possess feature A and “some embodiments” possess feature B should be interpreted as an express indication that the inventors also contemplate embodiments which combine features A and B (unless the description states otherwise or features A and B are fundamentally incompatible). This is the case even if features A and B are illustrated in different drawings and/or mentioned in different paragraphs, sections or sentences.

[0329] It is therefore intended that the following appended claims and claims hereafter introduced are interpreted to include all such modifications, permutations, additions, omissions, and sub-combinations as may reasonably be inferred. The scope of the claims should not be limited by the preferred embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. A fiber optic imaging system for generating multipath image data, the system comprising: an optical coherence tomography (OCT) system comprising an OCT fiber optically connected to an interferometric detector; a multi-clad fiber (MCF) for receiving sampled light that has interacted with a sample and propagating the sampled light as a fundamental MCF mode and one or more higher order MCF modes; an optical joint for coupling the sampled light from the MCF into the OCT fiber, the optical joint configured to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber where the at least some light energy propagates as OCT return light.
2. The imaging system of claim 1 comprising: a controller connected to the interferometric detector to receive, from the interferometric detector, OCT image data based on the OCT return light, the controller configured to: extract fundamental mode image data and higher order mode image data from the OCT image data, wherein the extraction is based on A-line coordinates of the OCT image data; and generate multipath image data (e.g. diagnostic image data) based on both of the fundamental mode image data and the higher order mode image data.
3. The imaging system of claim 2 wherein the controller is configured to: extract the fundamental mode image data based on light energy from the fundamental MCF mode that is coupled into the fundamental mode of the OCT fiber; and extract the higher order mode image data based on light energy from the one or more higher order MCF modes that is coupled into the fundamental mode of the OCT fiber.
4. The imaging system of claim 1 wherein the controller is configured to extract the fundamental mode image data and the higher order mode image data from the OCT image data using upper and lower A-line coordinate thresholds.

5. The imaging system of claim 1 wherein the MCF introduces a phase delay between the fundamental MCF mode and the one or more higher order MCF modes.
6. The imaging system of claim 5 wherein the controller is configured to extract the fundamental mode image data and the higher order mode image data based on the phase delay between the fundamental MCF mode and the one or more higher order MCF modes.
7. The imaging system of claim 5 wherein an amount of the phase delay depends on a length of the MCF and wherein the amount of the phase delay is sufficient to enable extraction of the fundamental mode image data and the higher order mode image data from the OCT image data based on the A-line coordinates of the OCT image data.
8. The imaging system of claim 1 wherein the MCF comprises a MCF core and at least one light-transmitting MCF cladding and wherein the fundamental MCF mode propagates substantially in the MCF core and the one or more higher order MCF modes propagate substantially in the at least one MCF cladding.
9. The imaging system of claim 1 wherein the fundamental MCF mode comprises light energy received from relatively low numerical aperture portions of the sampled light and the one or more higher order MCF modes comprise light energy received from relatively high numerical aperture portions of the sampled light.
10. The imaging system of claim 1 wherein the MCF propagates incident light from the optical joint toward the sample in an incident direction opposite to a direction of propagation of the sampled light, at least a portion of the incident light interacting with the sample to become the sampled light.
11. The imaging system of claim 10 wherein the incident light comprises: fundamental mode incident light that propagates in the incident direction in a fundamental mode of the MCF; and higher order mode incident light that propagates in the incident direction in one or more higher order modes of the MCF.
12. The imaging system of claim 11 wherein the one or more higher order MCF modes of the sampled light comprise a first component corresponding to the fundamental mode incident light that has interacted with the sample and a second component corresponding to the higher order mode incident light that has interacted with the sample.
13. The imaging system of claim 12 wherein the controller is configured to extract the higher order mode image data based on light energy from the first component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.
14. The imaging system of claim 13 wherein the controller is configured to extract additional higher order mode image data from the OCT image data based on light energy from the second component of the one or more higher order MCF modes of the sampled light that is coupled into the fundamental mode of the OCT fiber.
15. The imaging system of claim 14 wherein the controller is configured to extract the additional higher order mode image data based on A-line coordinates of the OCT image data.
16. The imaging system of claim 14 wherein the controller is configured to extract the fundamental mode image data, the higher order mode image data and the additional higher order mode image data based on differential phase delay between different propagation modes introduced in the MCF.
17. The imaging system of claim 14 wherein the controller is configured to generate multipath image data (e.g. diagnostic image data) based at least in part on the additional higher order mode image data.
18. The imaging system of claim 1 wherein the optical joint is configured to couple at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into the fundamental of the OCT fiber by introducing an offset between the MCF and the OCT fiber.
19. The imaging system of claim 18 wherein the offset comprises a spatial offset between an axis of the MCF and an axis of the OCT fiber.
20. A method for generating multipath image data, the method comprising: providing a multi-clad fiber (MCF); receiving sampled light that has interacted with the sample and propagating the sampled light in the MCF as a fundamental MCF mode and one or more higher order MCF modes; coupling the sampled light from the MCF into an optical coherence tomography (OCT) fiber, wherein coupling the sampled light comprises coupling at least some light energy from both the fundamental MCF mode and the one or more higher order MCF modes into a fundamental mode of the OCT fiber, where the at least some light energy propagates as OCT return light; generating OCT image data based on the OCT return light; extracting fundamental mode image data and higher order mode image data from the OCT image data, wherein the extracting is based on A-line coordinates of the OCT image data; and generating the multipath image data based on both of the fundamental mode image data and the higher order mode image data.
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